

Stochastic Modelling of NIFTY BANK Index using Monte Carlo Methods



CHILIVARY VISHAL

Integrated M.Sc Mathematics, National Institute of Technology Warangal

ABSTRACT

This study examines the stochastic behaviour and risk characteristics of the NIFTY Bank Index using simulation-based methods applied to historical daily data from January 2015 to January 2026. The analysis is based on logarithmic returns, and exploratory examination reveals non-normality, volatility clustering, and extreme observations, indicating limitations of standard assumptions. To address this, both a parametric Geometric Brownian Motion model and a non-parametric bootstrap approach are employed. The former estimates drift and volatility from historical data, while the latter generates return sequences directly from the empirical distribution through resampling.

Monte Carlo simulation is used to obtain future price trajectories over a one-year horizon, and risk is quantified using Value at Risk and Conditional Value at Risk. The results show that the Geometric Brownian Motion model produces moderate risk estimates due to its normality assumption, whereas the bootstrap method captures heavy tails and yields higher, more realistic risk measures. These findings emphasize the importance of distributional assumptions and the role of empirical methods in accurate financial risk assessment.

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List of Abbreviations

Symbol / Abbreviation	Meaning
GBM	Geometric Brownian Motion
EDA	Exploratory Data Analysis
VaR	Value at Risk
CVaR	Conditional Value at Risk
NSE	National Stock Exchange
PDF	Probability Density Function
CDF	Cumulative Distribution Function
ECDF	Empirical Cumulative Distribution Function
INR	Indian Rupee
SDE	Stochastic Differential Equation
$S(t)$	Index (stock) price at time t
$S(0)$	Initial index price
$R(t)$	Return at time t
r_t	Logarithmic return
μ	Drift (mean return)
σ	Volatility (standard deviation of returns)
$B(t)$	Brownian Motion
$dB(t)$	Increment of Brownian Motion
dt	Infinitesimal time increment
$\xi(t)$	Standard normal random variable $N(0,1)$
n	Time horizon
N	Number of time steps
M	Number of simulation paths
α	Confidence level
$\mathbb{E}$	Expectation operator
P	Probability measure
Δt	Discrete time step

Chapter 1

Introduction

1.1 Background

Financial markets exhibit continuous fluctuations driven by a combination of systematic trends and random external influences such as economic news, policy changes, and investor behaviour. These fluctuations make asset prices inherently uncertain, motivating the use of probabilistic and stochastic models rather than deterministic frameworks. From a mathematical perspective, the primary interest lies in understanding the distributional behaviour of prices and returns, rather than predicting a single future outcome.

Stochastic processes have therefore become central to financial modelling. In particular, Brownian motion provides a natural mathematical representation of random price movements, and its extensions form the foundation of many classical models in financial mathematics. Since asset prices are non-stationary and strictly positive, modelling is typically performed on logarithmic returns, which are more suitable for statistical analysis and probabilistic interpretation.

One of the most widely used continuous-time models for asset prices is the Geometric Brownian Motion (GBM), which combines deterministic drift with random fluctuations driven by Brownian motion. Despite its simplifying assumptions, GBM remains a benchmark model due to its mathematical tractability and interpretability. When analytical solutions are unavailable or when distributional properties are of interest, Monte Carlo simulation serves as an effective numerical tool by generating a large ensemble of possible future paths.

In recent years, simulation-based approaches have gained importance not only for forecasting but also for risk analysis, particularly in understanding uncertainty and extreme outcomes. This background motivates the use of stochastic modelling and Monte Carlo methods to study financial indices such as the NIFTY BANK index from a probabilistic viewpoint.

1.2 Motivation

Traditional approaches to financial analysis often focus on point forecasts or deterministic trends, which provide limited insight into the uncertainty and risk inherent in financial markets. In reality, market behaviour is dominated by randomness, and small fluctuations can accumulate over time to produce significantly different outcomes. This makes simulation-based and probabilistic methods more appropriate for analysing financial time series.

The NIFTY BANK index plays a crucial role in the Indian equity market and reflects the collective behaviour of major banking institutions. Its long historical record and exposure to macroeconomic events make it an ideal candidate for studying stochastic dynamics and market risk. Rather than attempting short-term prediction, analysing the distribution of possible future outcomes offers a more meaningful understanding of market uncertainty.

While Geometric Brownian Motion provides a simple and interpretable stochastic framework, its assumptions of constant volatility and normally distributed returns are often inconsistent with empirical data. Financial returns frequently exhibit heavy tails and extreme movements, especially during periods of market stress. This motivates the comparison of parametric models with non-parametric approaches that rely directly on historical data.

The motivation of this project is therefore to examine how different modelling assumptions influence simulated outcomes and risk estimates. By combining classical stochastic modelling

with data-driven bootstrap techniques, the study aims to provide a balanced and realistic assessment of uncertainty in the evolution of the NIFTY BANK index.

1.3 Problem Statement

This project focusses on modelling the future evolution and risk profile of the NIFTY BANK index using stochastic simulation techniques. A Geometric Brownian Motion based Monte Carlo framework is employed as a parametric baseline, while a non-parametric bootstrap approach is used to retain empirical return characteristics.

The problem addressed is to examine how these two modelling approaches differ in their representation of uncertainty and downside risk, particularly in the estimation of Value at Risk and Conditional Value at Risk. By comparing the resulting simulated distributions, the study evaluates the impact of modelling assumptions on risk assessment for a major Indian financial index.

1.4 Objectives

The specific objectives of this project are as follows:

1. To analyse the historical behaviour of the NIFTY BANK index through return-based statistical and exploratory analysis.
2. To model the index dynamics using a Geometric Brownian Motion framework with parameters estimated from historical data.
3. To implement Monte Carlo simulation for generating future index paths and studying the associated uncertainty.
4. To develop a non-parametric bootstrap simulation approach that preserves empirical return distributions.

5. To estimate downside risk using simulation-based measures such as Value at Risk (VaR) and Conditional Value at Risk (CVaR).
6. To compare parametric and non-parametric simulation methods and assess the impact of modelling assumptions on risk estimation.

1.5 Scope of Study

This study focuses on the stochastic modelling and simulation of the NIFTY Bank Index using probabilistic techniques, with particular emphasis on Monte Carlo methods. The analysis is based on historical daily closing price data of the index over a specified time period, which is used to estimate model parameters such as drift and volatility.

The modelling framework primarily relies on the assumption that the index follows a Geometric Brownian Motion (GBM), which enables the simulation of future price paths under a continuous-time stochastic process. In addition to the parametric GBM-based approach, a non-parametric bootstrap method is also employed to capture empirical distributional characteristics of returns without imposing strict assumptions on their underlying distribution.

The scope of the study includes exploratory data analysis to understand statistical properties such as distribution of returns, volatility behaviour, and temporal dynamics. Further, Monte Carlo simulation is used to generate multiple potential future trajectories of the index, which are then analysed to estimate risk measures such as Value at Risk (VaR) and Conditional Value at Risk (CVaR).

However, the study is limited to modelling based on historical price data and does not incorporate external macroeconomic factors, market microstructure effects, or regime-switching behaviour. The assumption of constant drift and volatility in the GBM framework may not fully capture real market complexities such as volatility clustering and sudden market shocks. Additionally, the analysis is confined to discrete-time simulation and does not account for continuous-time trading or transaction costs.

Despite these limitations, the study provides a structured probabilistic framework for understanding market uncertainty and risk, and demonstrates the applicability of stochastic modelling techniques in financial analysis.

Chapter 2

Literature Review

This chapter presents a critical review of existing literature relevant to stochastic modelling of financial time series, with particular emphasis on Brownian motion, Geometric Brownian Motion, and Monte Carlo simulation techniques. The origins of stochastic modelling in finance can be traced to the early nineteenth century with Robert Brown's observation of the random motion of pollen particles suspended in water, a phenomenon later termed Brownian motion. This physical interpretation of randomness was formalized mathematically by Norbert Wiener, providing a rigorous probabilistic framework for continuous-time random processes. Building on this foundation, Louis Bachelier (1900) first applied Brownian motion to financial markets, proposing that asset price fluctuations could be modelled as a random process driven by unpredictable future information. Although Bachelier's arithmetic Brownian motion allowed prices to become negative, his work established the conceptual foundation for modern mathematical finance and introduced the idea of modelling markets probabilistically rather than deterministically.

Subsequent refinements addressed the limitations of Bachelier's model. Samuelson proposed that asset prices follow a lognormal distribution, leading to the formulation of Geometric Brownian Motion (GBM), which ensures strictly positive prices and incorporate both drift and volatility. This model became central to financial theory and underpins the Black-Scholes framework for option pricing. Protonotarios and Tassopoulos (2021) provide a rigorous

mathematical exposition of Brownian motion and its role in financial modelling, demonstrating how stochastic differential equations and Itô calculus naturally arise when modelling log-price dynamics. Their numerical illustration shows that while Brownian-driven models capture a meaningful portion of real market variability, they remain idealized representations of complex financial behaviour.

With the development of GBM, Monte Carlo simulation emerged as a practical numerical tool for analysing stochastic models. Boyle first introduced Monte Carlo methods into finance, and later studies established simulation as a flexible framework for evaluating expectations, uncertainty, and risk when closed-form solutions are unavailable. Nwozo and Fadugba (2012) demonstrate the effectiveness of Monte Carlo simulation in pricing path-dependent options such as Asian and lookback options, where payoffs depend on the entire price trajectory rather than terminal values. Their work highlights Monte Carlo simulation as a versatile method for handling complex stochastic systems, albeit at the cost of increased computational effort.

Beyond derivative pricing, GBM-based Monte Carlo methods have been widely applied to stock price forecasting, particularly in emerging markets. Antwi (2017) applies a GBM - Monte Carlo framework to stocks listed on the Ghana Stock Exchange, evaluating model performance using Mean Absolute Percentage Error (MAPE) and reporting reasonable average forecasting accuracy. Similarly, Parungrojrat and Kidsom (2019) compare GBM and Monte Carlo techniques on SET50 stocks in Thailand, finding that both methods exhibit limited accuracy for long-horizon predictions and that forecast reliability deteriorates over time. More recently, Miah (2025) applies GBM-based Monte Carlo simulation to stocks listed on the Dhaka Stock Exchange, reporting low MAPE values for short-term forecasts. Collectively, these studies demonstrate the practical appeal of GBM for forecasting while also revealing its sensitivity to modelling assumptions and its limited ability to capture extreme market movements.

While forecasting studies focus primarily on point accuracy, other strands of literature emphasize risk and uncertainty. Luoma (2010) investigates the implications of transaction costs in option hedging under the Black-Scholes framework, using Monte Carlo simulations to compare alternative hedging strategies. The study shows that continuous rebalancing is impractical in real markets and that adaptive, move-based strategies outperform classical delta hedging when transaction costs are present. This work highlights the importance of distributional outcomes and risk-return trade-offs rather than average predictions alone.

Taken together, the literature establishes a progression from the physical origins of Brownian motion to its formal application in financial modelling, the development of Geometric Brownian Motion as a standard framework for asset prices, and the widespread adoption of Monte Carlo simulation for evaluating uncertainty and risk. Despite their mathematical elegance and analytical tractability, GBM-based models are often limited in capturing empirical features such as heavy tails, volatility clustering, and extreme market events. These limitations have motivated the use of simulation-based risk measures and data-driven approaches that better reflect real market behaviour. In this context, the present study aims to apply stochastic modelling techniques to the NIFTY Bank Index, with a particular focus on uncertainty quantification and downside risk analysis using both parametric and non-parametric simulation methods.

Chapter 3

Report on Present Investigation

3.1 Project Flow Framework

This section describes the overall methodological framework adopted in the present study for modelling and analysing the stochastic behaviour of the NIFTY Bank Index. The workflow is designed to systematically transform raw historical data into meaningful probabilistic insights through a combination of statistical analysis, stochastic modelling, and simulation-based risk assessment.

The study begins with the collection of historical daily closing price data of the NIFTY Bank Index. This raw dataset is subjected to preprocessing and transformation procedures, which include data cleaning, handling of missing values (if any), and conversion of price series into logarithmic returns. The transformation to log returns is essential as it stabilizes variance and enables the application of continuous-time stochastic models.

Following preprocessing, an exploratory data analysis (EDA) is performed to understand the statistical characteristics of the return series. This includes examining the distribution of returns, identifying volatility behaviour, and analysing temporal patterns present in the data. The insights obtained from EDA provide a foundation for selecting appropriate modelling approaches and highlight key empirical features such as non-normality and volatility clustering.

Based on these observations, the modelling framework is divided into two parallel approaches: a parametric and a non-parametric methodology. In the parametric framework, the index dynamics are modelled using Geometric Brownian Motion (GBM), which assumes that returns follow a normal distribution characterized by constant drift and volatility. The parameters of the model, namely drift and volatility, are estimated from historical log returns using their sample mean and standard deviation. These estimated parameters are then used to simulate future index price paths through Monte Carlo simulation, generating multiple possible realizations of the stochastic process.

In parallel, a non-parametric approach is implemented using bootstrap resampling techniques. Instead of assuming any specific distribution, this method relies directly on the empirical distribution of historical returns. By resampling observed returns with replacement, bootstrap simulation preserves key distributional features such as skewness, kurtosis, and extreme events, thereby providing an alternative representation of market behaviour. Simulated return paths generated through this approach are subsequently converted into corresponding price paths.

The outputs from both modelling approaches consist of simulated price and return distributions over the chosen time horizon. These simulated outcomes are then analysed to quantify financial risk using measures such as Value at Risk (VaR) and Conditional Value at Risk (CVaR), which capture the potential magnitude of losses under adverse market conditions.

A comparative analysis is performed between the parametric (GBM-based) and non-parametric (bootstrap-based) models to evaluate their effectiveness in capturing uncertainty and extreme risk. Further, a sensitivity analysis is conducted to assess the impact of underlying assumptions on the simulation results.

Finally, the study synthesizes the findings to derive meaningful insights regarding the stochastic behaviour of the NIFTY Bank Index and the associated risk characteristics. The overall framework thus integrates data-driven analysis with stochastic modelling techniques to provide a comprehensive understanding of market uncertainty.

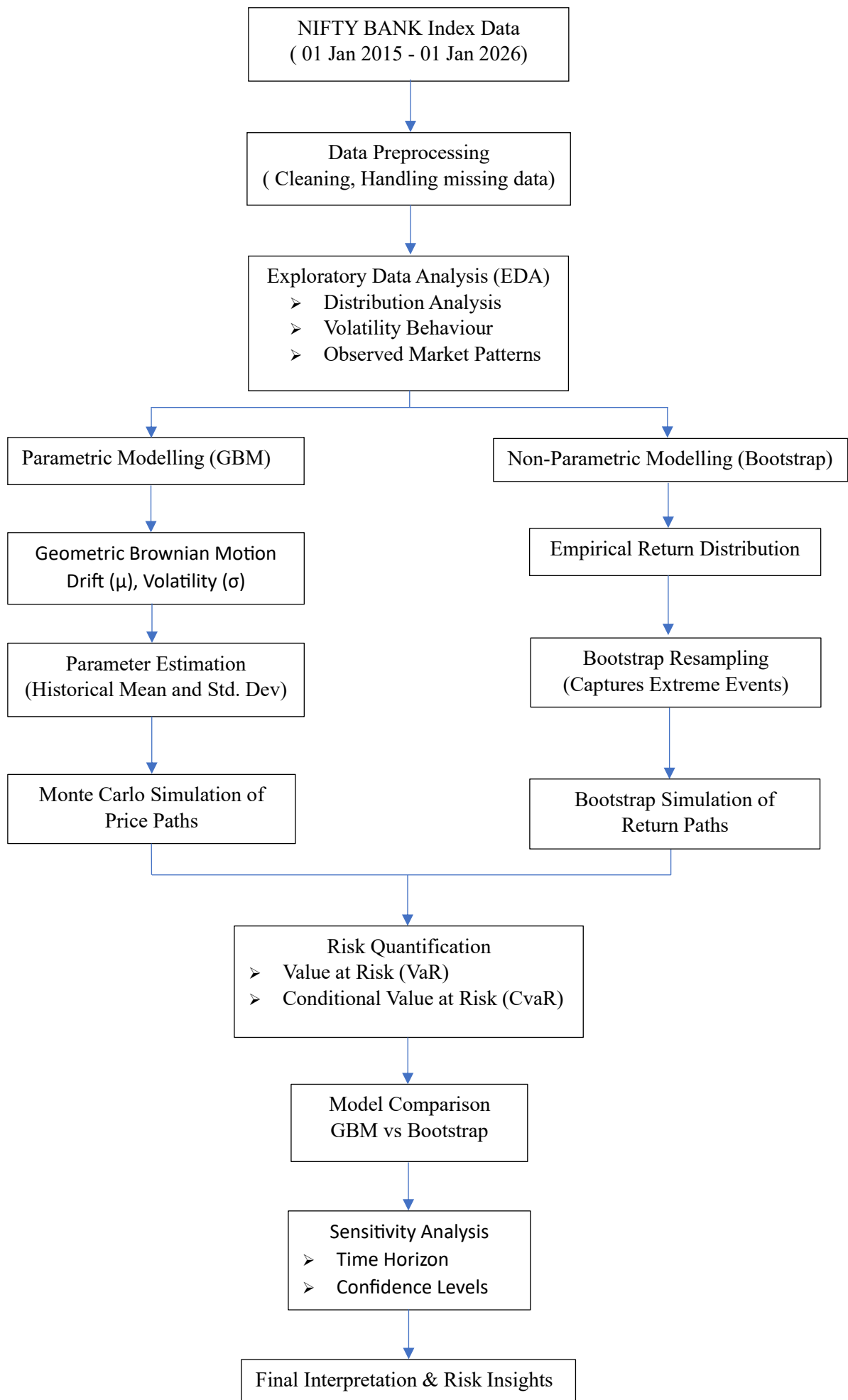


Figure 3.1: Project Flow

3.2 Dataset Description and Data Source

The NIFTY BANK Index is a sectoral stock market index that tracks the performance of the most liquid and large-cap banking stocks listed on the National Stock Exchange (NSE) of India, and it currently comprises 14 major banking institutions.

Table 3.1: NIFTY BANK Index List as on 11/02/2026

S. No	Bank Name	S. No	Bank Name
1	HDFC Bank	8	Union Bank
2	State Bank of India	9	Canara Bank
3	ICICI Bank	10	AU Small Finance
4	Kotak Mahindra Bank	11	IndusInd Bank
5	Axis Bank	12	Federal Bank
6	Bank of Baroda	13	IDFC First Bank
7	Punjab National Bank	14	Yes Bank

The dataset used in this project consists of daily historical data of the NIFTY BANK Index covering the period from 01/01/2015 to 01/01/2026. After cleaning and preprocessing, the dataset contains 2726 trading-day observations, representing 11 years of market activity, including periods of economic expansion, market corrections, and high-volatility episodes.

Table 3.2: First five rows of the dataset

	Date	Open	High	Low	Close	Shares Traded	Turnover (INR Cr)
0	2015-01-01	18728.20	18781.55	18638.85	18750.45	20469847.0	810.17
1	2015-01-02	18752.20	19118.85	18752.20	19057.80	41288039.0	1713.48
2	2015-01-05	19155.20	19166.00	18987.70	19017.40	36318747.0	1497.40
3	2015-01-06	18874.60	18874.60	18388.35	18430.75	52069776.0	2124.64
4	2015-01-07	18382.55	18482.05	18211.50	18304.25	58241387.0	2341.58

Dataset Characteristics:

- Frequency: Daily (trading days only; weekends and market holidays excluded).
- Time Span: 01 Jan 2015 - 01 Jan 2026.
- Number of Trading Days: 2726.
- Index Type: Sectoral equity index (Banking sector).
- Market: Indian equity market (NSE).

Available Variables:

The dataset includes the following fields:

- Date: Trading date.
- Open: Opening index value.
- High: Highest index value during the trading day.
- Low: Lowest index value during the trading day.
- Close: Closing index value.
- Shares Traded: Total trading volume.
- Turnover (INR Cr): Daily turnover in crore rupees.

Key Observations:

- The data is already recorded on trading days only, eliminating the need for artificial interpolation.
- The closing price series exhibits non-stationarity, motivating the use of return-based modelling.
- Daily log returns display volatility clustering, skewness, and excess kurtosis, indicating deviations from normality.
- These empirical characteristics justify the use of both:
 - Parametric Monte Carlo simulation (GBM), and
 - Non-parametric bootstrap simulation, which preserves the empirical return distribution.

This dataset provides a sufficiently long and rich historical record to estimate model parameters, perform simulation-based forecasting, and analyse downside risk using Monte Carlo methods.

Data Source: The dataset used in this project was obtained from the official website of the National Stock Exchange (NSE) of India.

3.3 Data Preprocessing

The dataset was first converted into a consistent datetime format and sorted chronologically to preserve the correct time sequence. Numerical columns were checked for missing or invalid values, and any rows with incomplete data were removed.

Daily returns and logarithmic returns were computed from the closing price series to transform the non-stationary price data into a suitable form for stochastic modelling. This cleaned and transformed dataset formed the basis for subsequent exploratory analysis and Monte Carlo simulation.

3.4 Exploratory Data Analysis (EDA)

This section examines the statistical and structural characteristics of the NIFTY Bank Index data to gain preliminary insights into its behaviour. The analysis includes the study of price distributions, return dynamics, trading activity, and volatility patterns. Particular emphasis is placed on identifying features such as non-normality, variability, and temporal dependencies. These observations provide the basis for selecting appropriate stochastic models in the subsequent sections.

3.4.1 Descriptive Statistics and Distribution Analysis

This subsection summarizes the key statistical properties of the dataset, including measures of central tendency and dispersion. It also examines the distribution of price variables and returns through graphical analysis to identify patterns such as spread, skewness, and overall data behaviour.

Table 3.3: Statistical Analysis of the Dataset

	Open	High	Low	Close	Shares Traded	Turnover (INR Cr)
count	2726.0000	2726.0000	2726.0000	2726.0000	2.718000e+03	2718.0000
mean	32760.0274	32977.4161	32506.9280	32745.3336	1.970816e+08	6263.0999
min	13715.1000	13844.4500	13407.2500	13555.7000	8.938951e+06	267.0900
25%	22362.8750	22495.2250	22066.8625	22271.7250	1.013737e+08	3167.8525
50%	30535.5000	30700.6250	30238.9500	30450.5250	1.611033e+08	5612.6350
75%	43012.6375	43296.4375	42766.8375	42997.2500	2.502116e+08	8148.9650
max	60102.0500	60114.3000	59598.9500	59777.2000	1.568143e+09	48724.2500
std	12256.2041	12304.7941	12218.7390	12264.9013	1.401957e+08	4022.4505

Histograms

1. Date:

The histogram of observation dates shows how trading data are distributed across the study period from 2015 to 2026.

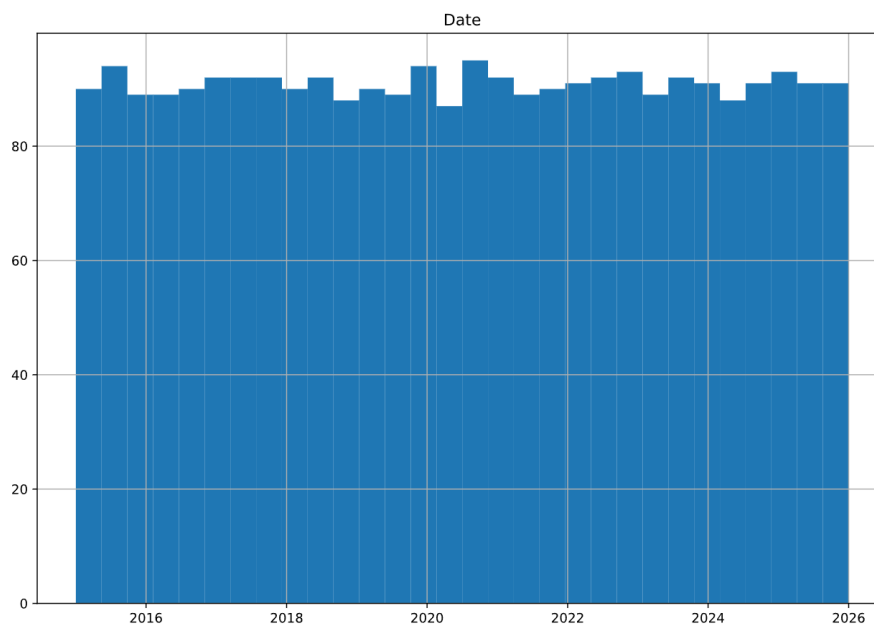


Figure 3.2: Histogram of Number of Trading dates

The nearly uniform bar heights indicate consistent data availability throughout the years, confirming the absence of significant gaps or uneven sampling. This balanced coverage ensures reliable estimation of model parameters based on continuous market observations.

2. Open Price:

The histogram of opening prices illustrates the distribution of the NIFTY BANK index levels over the study period.

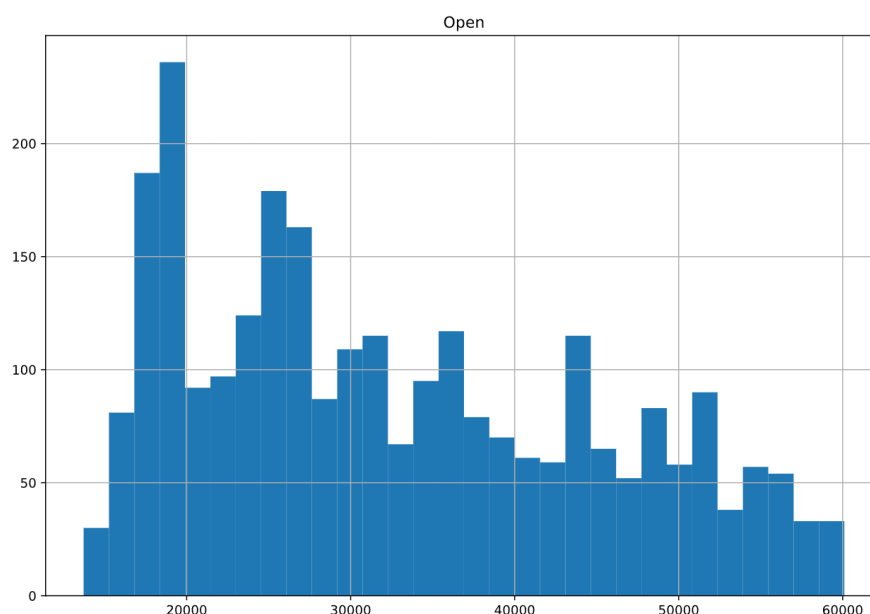


Figure 3.3: Histogram showing the distribution of NIFTY BANK opening prices during 2015-2026

The spread of values reflects the long-term upward movement of the index, with observations distributed across multiple price ranges rather than concentrated around a single level. The rightward extension of the distribution indicates higher index values in recent years, highlighting the non-stationary nature of price levels and motivating the use of return-based analysis for stochastic modelling.

3. High Price:

The histogram of daily high prices shows the range of peak index levels reached during trading sessions over the study period.

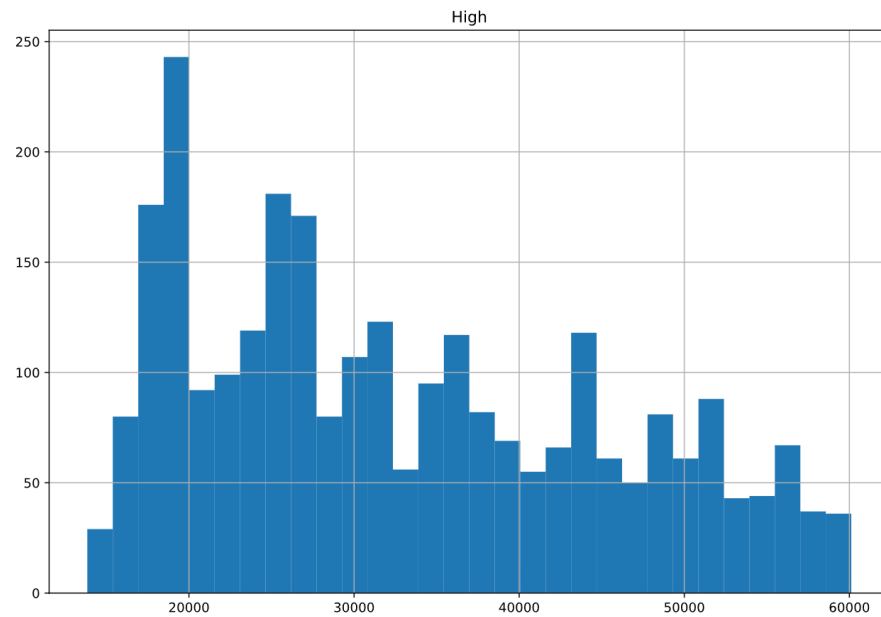


Figure 3.4: Histogram showing the distribution of daily high prices of the NIFTY BANK index

Similar to the opening price distribution, the values are spread across progressively higher price ranges, reflecting the long-term growth of the NIFTY BANK index. The wide dispersion further indicates variability in market conditions across different periods, reinforcing the non-stationary behaviour of price levels and supporting the transition to return-based modelling.

4. Low Price:

The histogram of daily low prices represents the range of minimum index levels recorded during trading sessions over the study period.

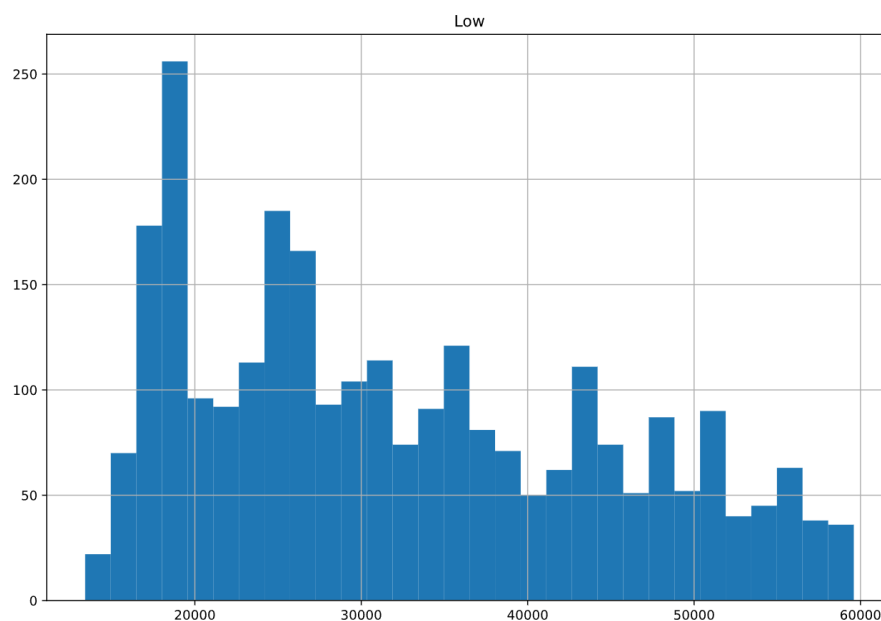


Figure 3.5: Histogram showing the distribution of daily low prices of the NIFTY BANK index

The distribution spans a wide interval of price levels, reflecting the gradual upward movement of the NIFTY BANK index along with varying market conditions across time. The dispersion of values further highlights the non-stationary nature of price series, supporting the need for return-based analysis in subsequent modelling.

5. Closing Price:

The histogram of closing prices shows a wide spread of index levels over the study period, reflecting the long-term upward movement of the NIFTY BANK index.

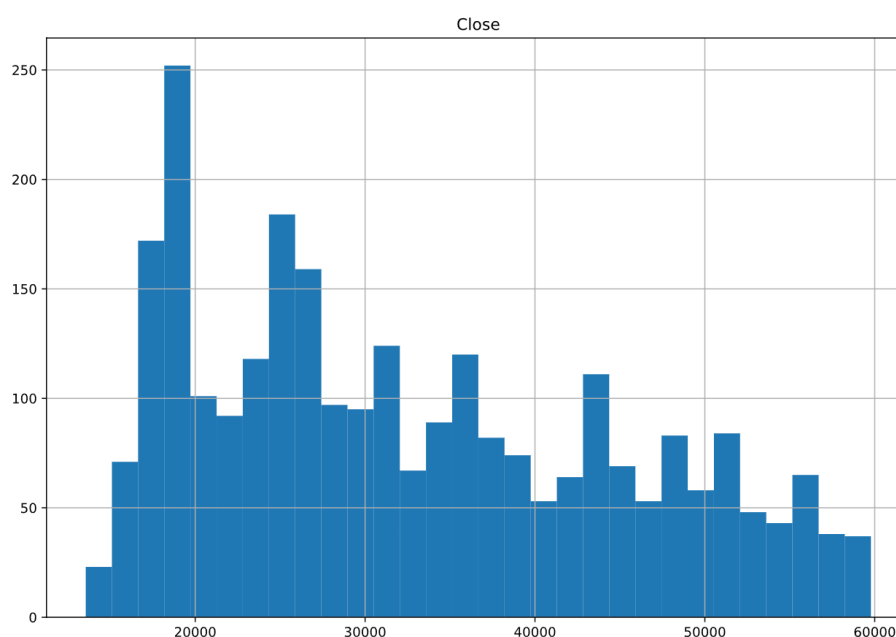


Figure 3.6: Histogram showing the distribution of daily closing prices of the NIFTY BANK index

The absence of concentration around a fixed level indicates non-stationary price behaviour, supporting the use of return-based analysis for modelling.

3.4.2 Price Dynamics

This subsection analyses the temporal behaviour of the NIFTY Bank Index through its price movements over time. It examines trends, fluctuations, and intraday variations to understand the non-stationary nature of the price series. These observations help in identifying underlying market patterns and justify the use of return-based stochastic modelling.

Historical Closing Price

The time series plot of the NIFTY BANK closing prices illustrates the long-term evolution of the index over the study period from 2015 to 2026.

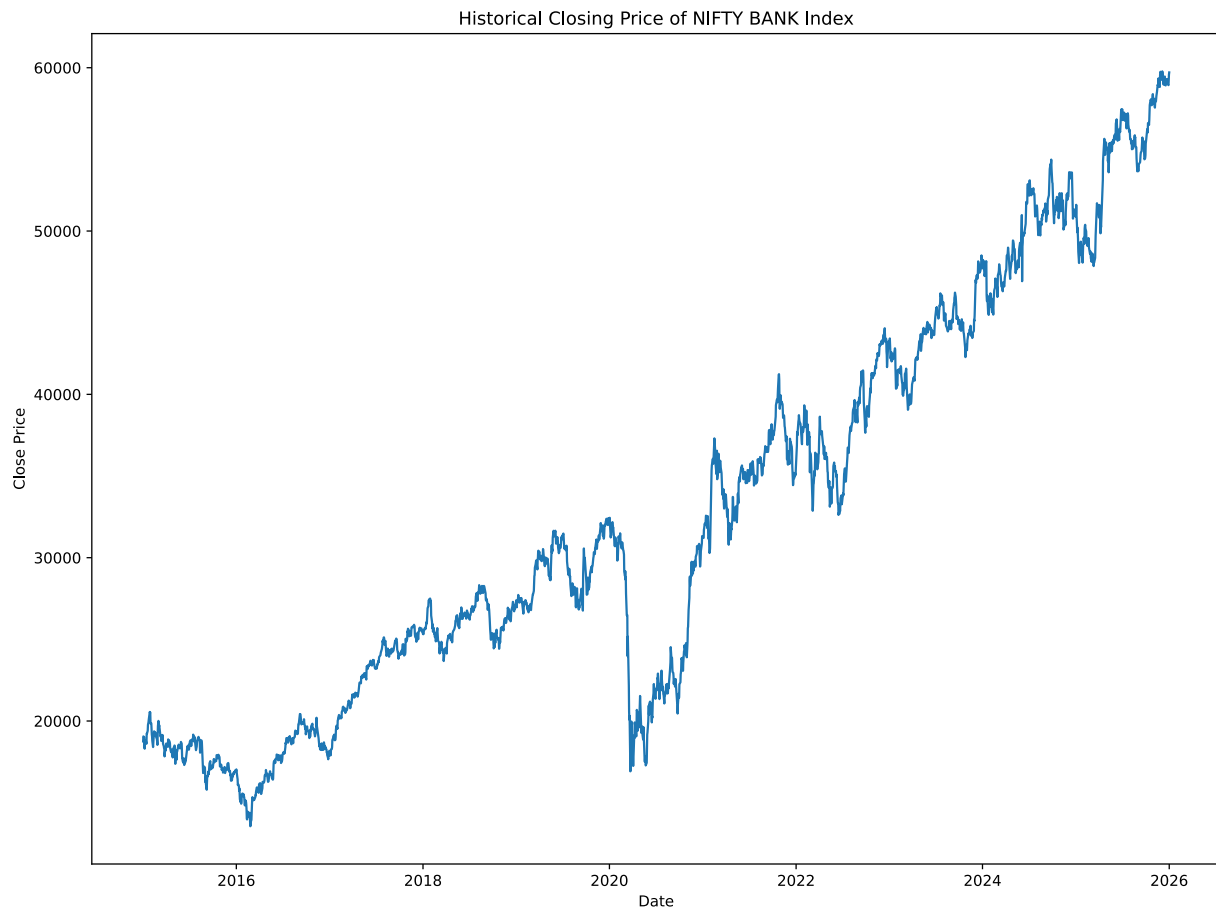


Figure 3.7: Historical closing price movement of the NIFTY BANK index

The series exhibits a clear upward trajectory, reflecting sustained growth in the banking sector over time, interspersed with noticeable periods of sharp declines and rapid recoveries. Significant fluctuations are visible during market stress phases, followed by renewed upward momentum, indicating changing market regimes and varying volatility conditions across years.

The presence of pronounced trends and changing variance suggests that the price series is non-stationary, with both mean and variability evolving over time. These characteristics highlight the stochastic nature of financial markets and justify transforming prices into logarithmic returns for subsequent modelling and simulation analysis.

Daily Price Range

The plot of the daily price range, defined by the difference between the high and low prices, illustrates fluctuations in intraday market volatility over the study period.

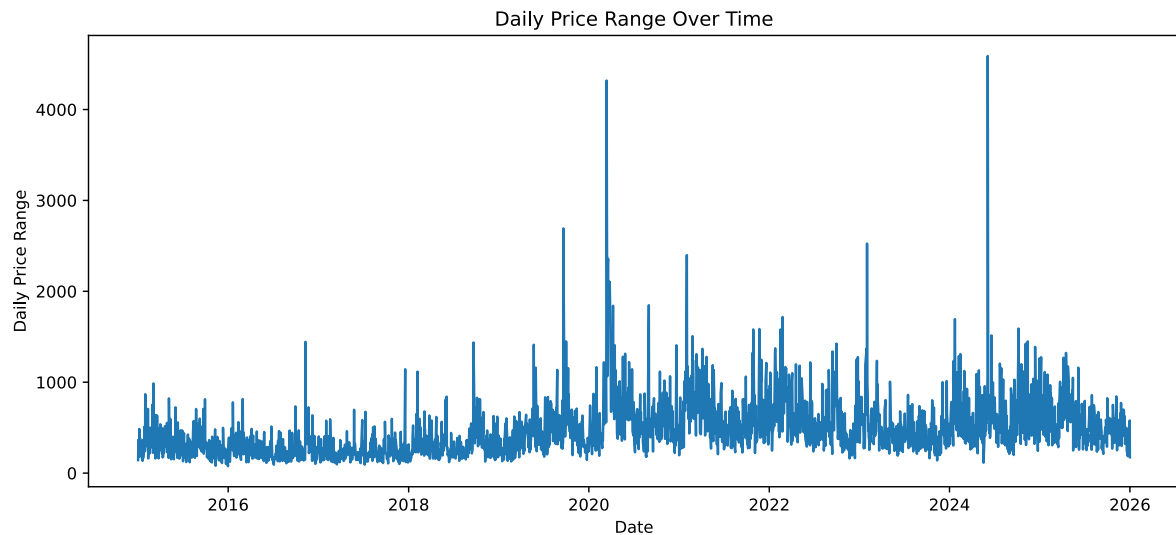


Figure 3.8: Daily price range (High-Low) of the NIFTY BANK index showing variations in intraday

Periods with larger ranges indicate heightened uncertainty and stronger trading activity, while narrower ranges correspond to relatively stable market conditions. The clustering of large movements suggests that volatility tends to occur in bursts rather than uniformly over time, reflecting a common characteristic of financial markets.

3.4.3 Trading Activity Analysis

This subsection examines the trading behaviour of the NIFTY Bank Index through measures such as shares traded and turnover. It analyses the distribution and temporal variation of trading volume to understand market participation and liquidity. These insights help relate trading activity to periods of heightened volatility and market movements.

Shares Traded

The histogram of shares traded shows a right-skewed distribution, with most trading volumes concentrated at lower levels and occasional periods of exceptionally high activity.

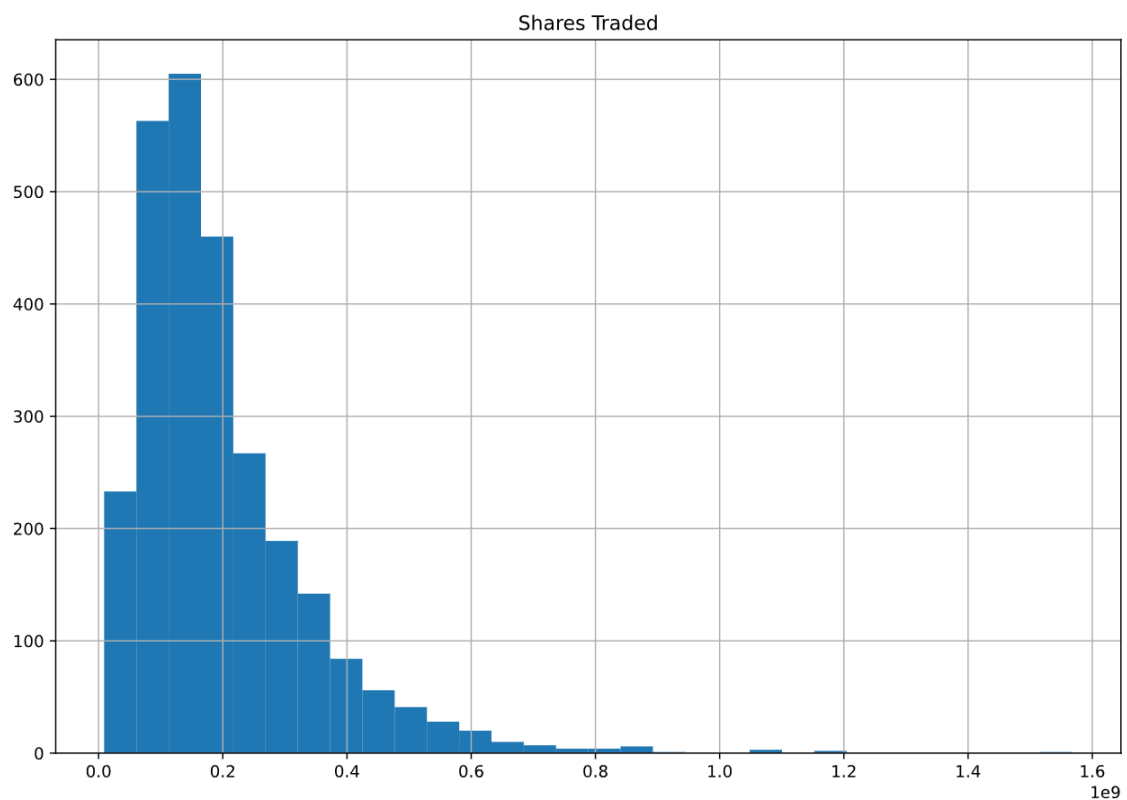


Figure 3.9: Distribution of daily shares traded for the NIFTY BANK index

This indicates the presence of intermittent market surges, often associated with heightened volatility or major market events.

Trading Volume Over Time

The time series of trading volume illustrates variations in market participation throughout the study period.

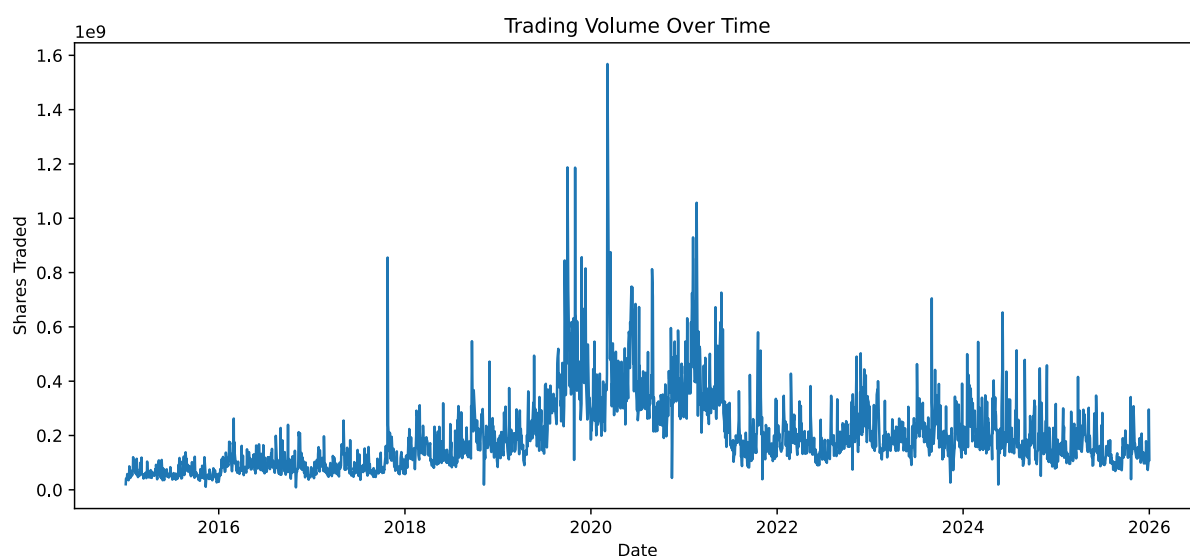


Figure 3.10: Daily trading volume of the NIFTY BANK index highlighting variations in market participation

The series shows intermittent spikes in trading activity, indicating periods of increased investor interest and heightened market reactions to economic or financial events. These bursts of activity are not uniformly distributed, suggesting that trading intensity varies across market conditions. Such fluctuations in volume often coincide with phases of higher price volatility, reflecting changing liquidity and market sentiment over time.

3.4.4 Return Distribution Analysis

This subsection focuses on the statistical behaviour of daily returns derived from the index. It examines the shape and spread of the return distribution to identify features such as concentration around the mean, skewness, and the presence of extreme values. These characteristics are important for assessing the suitability of probabilistic models.

Distribution of Daily Returns

The general daily return follows the below equation.

$$\text{Return } (R_t) = \frac{S_t - S_{t-1}}{S_{t-1}}$$

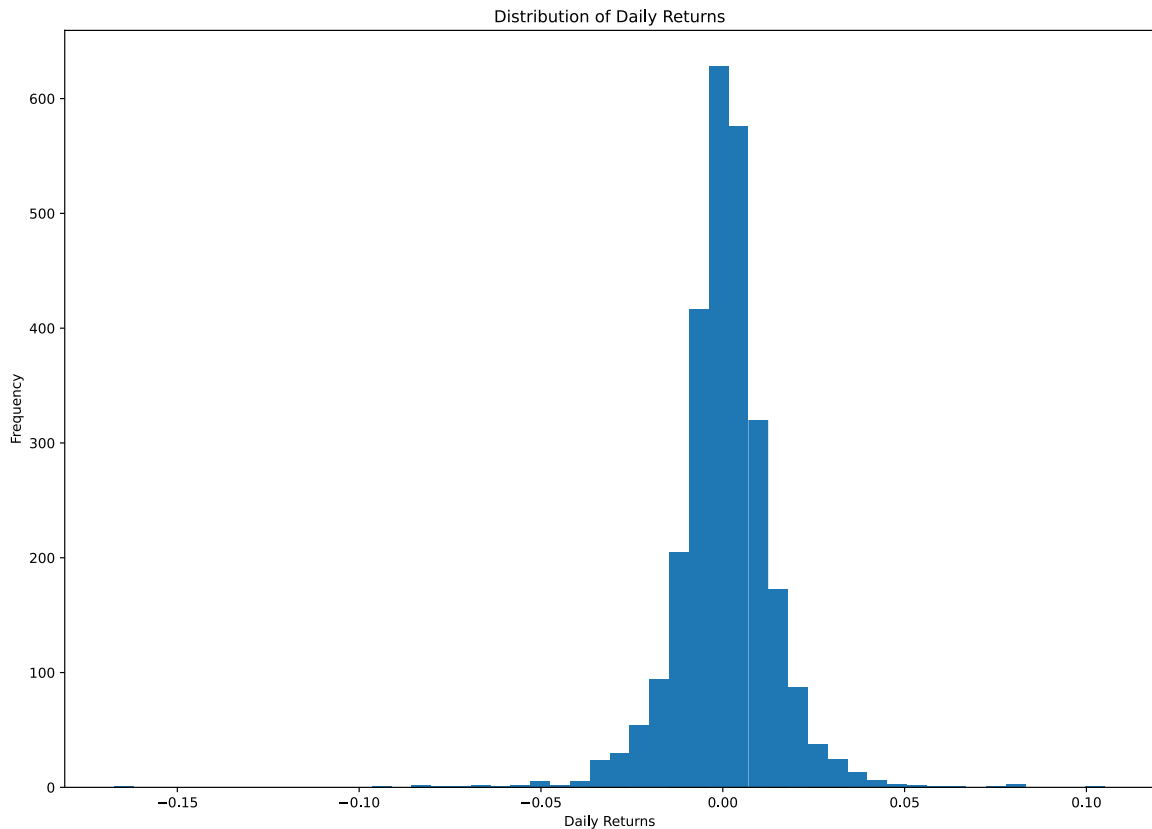


Figure 3.11: Distribution of Daily Returns

The distribution is sharply peaked around zero, indicating that most daily returns are small in magnitude, while large positive or negative returns occur less frequently.

3.4.5 Volatility Analysis

This subsection examines the variability of returns over time to understand the dynamic nature of market risk. It analyses rolling measures of volatility to identify periods of high and low fluctuations, highlighting the presence of volatility clustering. These observations provide insight into the time-varying behaviour of uncertainty in the index.

Rolling 30 - Day Volatility is the standard deviation of previous 30 daily returns, updated continuously through time.

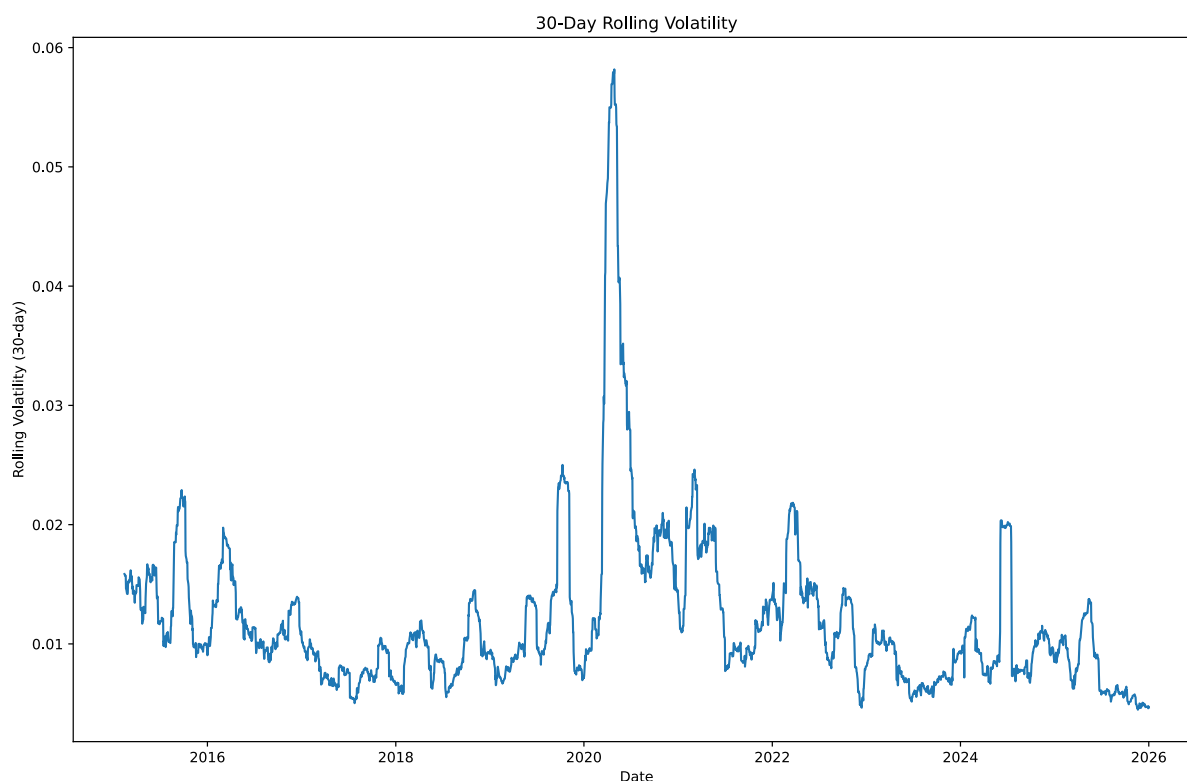


Figure 3.12: 30-day rolling volatility of the NIFTY BANK index showing time-varying market risk

The series highlights periods of changing market uncertainty, where spikes indicate phases of elevated volatility and increased price fluctuations, while lower values correspond to relatively stable market conditions. The clustering of high-volatility periods suggests that market risk varies over time rather than remaining constant, a key characteristic of financial return series and an important consideration for stochastic modelling.

3.4.6 Correlation Structure

This subsection analyses the relationships among the key variables of the dataset to understand their degree of interdependence. It examines the correlation between price components and trading activity to identify common trends and underlying patterns. These insights help justify the selection of relevant variables for subsequent modelling.

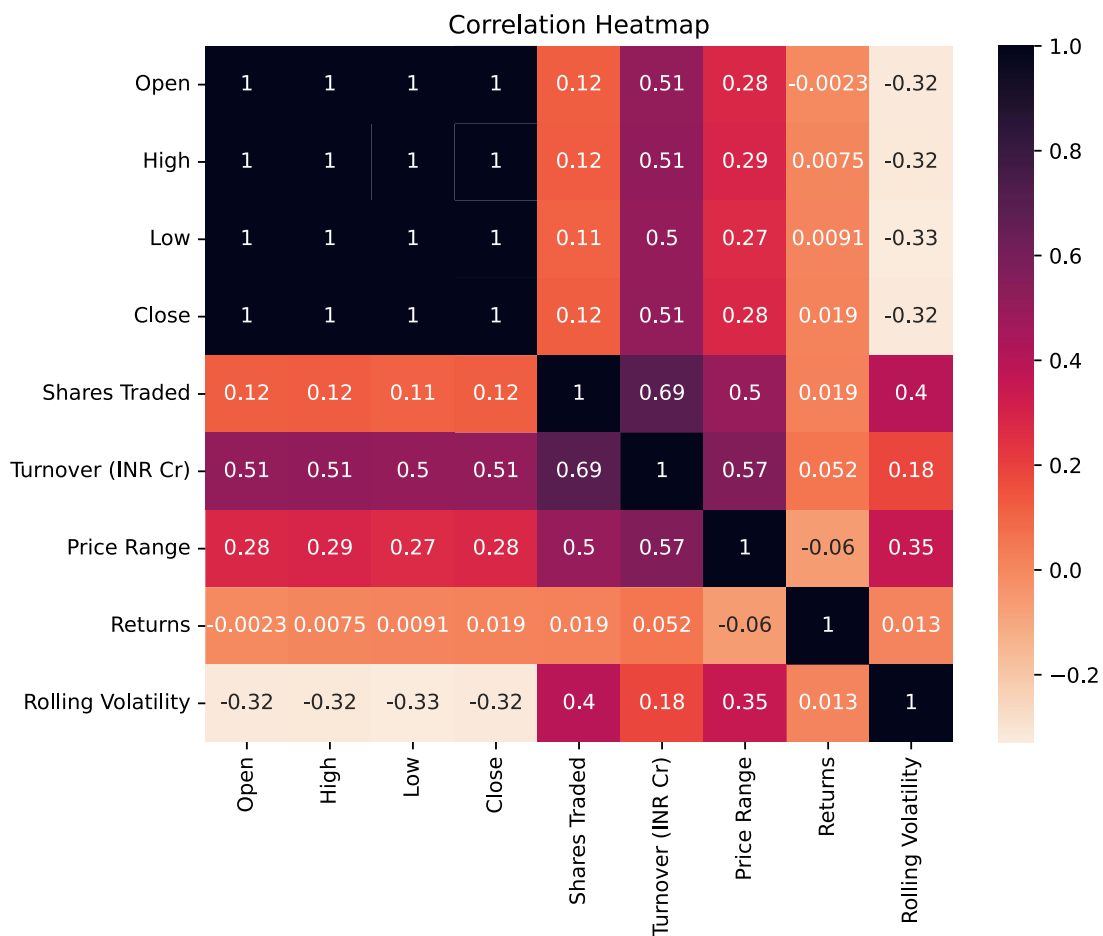


Figure 3.13: Correlation heatmap showing relationships among price variables of the NIFTY BANK index

A strong positive correlation is observed between the Open, High, Low, and Close prices, indicating that these variables move closely together as they represent different observations of the same underlying market level within a trading session. In contrast, trading volume shows comparatively weaker correlation with price variables, suggesting that changes in market participation do not always translate directly into proportional price movements.

This pattern highlights that price components largely reflect a common market trend, while trading activity captures a different dimension of market behaviour. The strong interdependence among price variables also justifies selecting a single representative measure, typically the closing price for return computation and subsequent stochastic modelling.

3.5 Stochastic Modelling Framework

This section develops the mathematical foundation for modelling the stochastic behaviour of the NIFTY Bank Index. It introduces the underlying probabilistic concepts and formulates both parametric and non-parametric approaches used to represent price dynamics. The section also outlines the estimation of model parameters and the construction of simulation-based price paths for further analysis.

3.5.1 The Brownian Motion Process

The Brownian motion $B(t)$ is used to capture the uncertainty in the future behaviour of a stochastic process and has the following properties.

- a. Independence of increments: $B(t) - B(s)$, for $t > s$, is independent of the past.
- b. Normal increments: $B(t) - B(s)$ has a Normal Distribution with mean 0 and variance $t - s$. If $s = 0$ then $B(t) - B(s) \sim N(0, t)$.
- c. Continuity of paths: $B(t), t \geq 0$ are continuous functions of t .

Louis Bachelier used Brownian motion to model the stock price. In a differential form, the Bachelier model can be written as,

$$dS(t) = \mu dt + \sigma dB(t) \quad (1)$$

Where $S(t)$ is the index price, $B(t)$ is the Brownian motion, μ is the return on the stock price, σ is the volatility of the stock price.

Equation 1 is called arithmetic Brownian motion.

Solution for equation 1 is:

$$\int_0^t dS(t) = \int_0^t \mu dt + \int_0^t \sigma dB(t)$$

$$S(t) - S(0) = \mu(t - 0) + \sigma(B(t) - B(0))$$

$$S(t) = S(0) + \mu(t - 0) + \sigma B(t) \quad (\because \text{at } t = 0, B(0) = 0)$$

The only weakness of the model is that stock prices can be negative which is contrary to reality.

3.5.2 Geometric Brownian Motion

This subsection introduces Geometric Brownian Motion as a continuous-time stochastic model for describing the evolution of index prices. It formulates the dynamics in terms of drift and volatility, capturing both deterministic growth and random fluctuations. The model serves as a fundamental basis for simulation and subsequent risk analysis.

Let $S(t)$ be the index price at time t . Then, the return on the index price is

$$\text{Return} = \frac{\text{Change in price}}{\text{Original price}}$$

Consider a small subsequent time interval $(t, t + \Delta t)$ during which $S(t)$ becomes

$$S(t + \Delta t) = S(t) + \Delta S(t)$$

The return on the index price between time t and $t + \Delta t$ is given by

$$\text{Return} = \frac{S(t + \Delta t) - S(t)}{S(t)} = \frac{\Delta S(t)}{S(t)}$$

This expression represents the instantaneous proportional change in the index price over the time interval Δt .

For sufficiently small time increments, the evolution of returns is assumed to consist of a deterministic component and a stochastic component.

The deterministic part accounts for the average rate of growth and is proportional to the elapsed time, given by $\mu\Delta t$, where μ : The mean of the historical returns (the drift parameter).

The stochastic component represents random market fluctuations and is modelled by standard Brownian motion $B(t)$. Under these assumptions, the proportional change in the index price may be expressed as,

$$\frac{\Delta S(t)}{S(t)} = \mu \Delta t + \sigma dB(t)$$

$$\frac{dS(t)}{S(t)} = \mu dt + \sigma dB(t)$$

or

$$dS(t) = \mu S(t) dt + \sigma S(t) dB(t) \quad (2)$$

Where σ (volatility): standard deviation of historical returns (measures the magnitude of randomness).

In integral form,

$$S(t) = \mu \int_0^t S(u) du + \sigma \int_0^t S(u) dB(u)$$

Brownian motion $B(t)$ is nowhere differentiable and fluctuates infinitely fast at every time scale. Due to these irregular paths, the classic integral is not well defined with respect to $B(t)$. Therefore, $B(t)$ is not integrable in the classical sense.

Here, $S(t)$ is said to be an Itô process.

An Itô process is used to represent the process of a price or portfolio value.

3.5.3 Analytic Solution to GBM

Let us assume f as a function of S (stock price) and t (time), df can be found by Taylor Expansion of $f(S, t)$ around $S(t)$ up to second order terms.

$$f(S(t) + \Delta S, t + \Delta t) \approx f(S(t), t) + \frac{\partial f}{\partial S} \Delta S + \frac{\partial f}{\partial t} \Delta t + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} (\Delta S)^2 + \frac{1}{2} \frac{\partial^2 f}{\partial t^2} (\Delta t)^2 + \frac{1}{2} \frac{\partial^2 f}{\partial t \partial S} \Delta t \Delta S$$

When Δt and ΔS are very small it can be written as:

$$df = \frac{\partial f}{\partial S} dS + \frac{\partial f}{\partial t} dt + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} (dS)^2 + \frac{1}{2} \frac{\partial^2 f}{\partial t^2} (dt)^2 + \frac{1}{2} \frac{\partial^2 f}{\partial t \partial S} dt dS$$

Using Itô rules, $(dt)^2 = 0$, $dt dB(t) = 0$, $(dB(t))^2 = dt$,

Since $dS(t) = \mu S(t) dt + \sigma S(t) dB(t)$, $dt dS(t) = 0$

We get
$$df = \frac{\partial f}{\partial S} dS + \frac{\partial f}{\partial t} dt + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} (dS)^2 \quad (3)$$

Let
$$f(S, t) = \ln(S)$$

i.e.,
$$f(S(t), t) = \ln(S(t))$$

where $f' = \frac{1}{S(t)}$, $f'' = -\frac{1}{S(t)^2}$

Applying it to equation (3),

$$d(\ln(S(t))) = \frac{1}{S} dS + \frac{1}{2} \left(-\frac{1}{S(t)^2} \right) (dS)^2 \quad (4)$$

From equation (2),

$$\begin{aligned} dS(t) &= \mu S(t) dt + \sigma S(t) dB(t) \\ (dS)^2 &= (\mu S(t) dt)^2 + (\sigma S(t) dB(t))^2 + 2(\mu S(t) dt)(\sigma S(t) dB(t)) \end{aligned}$$

Since $(dt)^2 = 0$, $dt dB(t) = 0$, $(dB(t))^2 = dt$

$$(dS)^2 = (\mu S(t) dt)^2 + (\sigma S(t) dB(t))^2 + 2(\mu S(t) dt)(\sigma S(t) dB(t))$$

$$(dS)^2 = \sigma^2 (S(t))^2 dt$$

Equation (4) becomes,

$$d(\ln(S(t))) = \frac{1}{S(t)} (\mu S(t) dt + \sigma S(t) dB(t)) + \frac{1}{2} \left(-\frac{1}{S(t)^2} \right) \sigma^2 (S(t))^2 dt$$

$$d(\ln(S(t))) = \frac{1}{S(t)} (\mu S(t) dt + \sigma S(t) dB(t)) - \frac{1}{2} \sigma^2 dt$$

$$d(\ln(S(t))) = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dB(t)$$

Integrating,

$$\ln(S(t)) = \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma B(t)$$

$$S(t) = S(0) * \exp \left(\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma B(t) \right) \quad (5)$$

3.5.4 Parameter Estimation

The implementation of the Geometric Brownian Motion (GBM) framework requires the estimation of the key parameters governing the stochastic dynamics of the index, namely the drift coefficient and volatility. These parameters are obtained from historical observations of the NIFTY Bank Index and form the basis for subsequent simulation.

Let $S(t)$ denote the closing price of the index at time t . The continuously compounded return over a discrete time interval is defined as:

$$r_i = \ln \left(\frac{S(t_i)}{S(t_{i-1})} \right)$$

This transformation converts the price series into a return series that is additive over time and suitable for stochastic modelling.

The drift parameter μ , representing the average rate of return, is estimated as the mean of the observed log returns:

$$\mu_i = \frac{1}{n} \sum_{i=1}^n \ln \left(\frac{S(t_i)}{S(t_{i-1})} \right)$$

Similarly, the volatility parameter σ , which captures the dispersion of returns, is estimated as the standard deviation of the log return series:

$$\sigma = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (r_i - \mu_i)^2}$$

This approach directly follows the empirical estimation of return and volatility from historical price data, where both quantities are derived from daily closing prices.

The estimation is carried out using daily observations, and the resulting parameters reflect the average behaviour of the index over the sample period. These estimated values of drift and volatility are then used as inputs for simulating future price paths under the GBM framework.

It is assumed that the estimated parameters remain constant over the simulation horizon. While this assumption simplifies the modelling process, it may not fully capture time-varying volatility and structural changes in the market. Nevertheless, this estimation method provides a practical and widely adopted basis for stochastic simulation of financial time series.

3.5.5 Simulation of Index Prices

Once the parameters of the GBM model are estimated, the future behaviour of the NIFTY Bank Index is simulated using a discrete-time approximation of the continuous stochastic process. This enables the generation of possible price trajectories that reflect both deterministic trends and random market fluctuations.

The GBM model is expressed in differential form as:

$$dS(t) = \mu S(t) dt + \sigma S(t) dB(t)$$

where $B(t)$ represents a standard Brownian motion capturing the random component of price evolution.

For computational purposes, this continuous-time model is discretized over small time intervals Δt . The corresponding discrete formulation can be written as:

$$\Delta S(t) = \mu S(t) \Delta t + \sigma S(t) \xi(t) \sqrt{\Delta t}$$

where:

- $\xi(t) \sim N(0,1)$ represents a standard normal random variable
- Δt is the time step (daily interval)

This discrete representation forms the basis for simulating stock price movements, where randomness is introduced through normally distributed shocks at each time step.

The simulation process begins with an initial value $S(0)$, taken as the most recent observed index value. At each subsequent time step, a random shock is generated and substituted into the discrete equation to compute the change in price. This process is repeated iteratively to obtain a complete price path over the desired time horizon.

To account for the inherent uncertainty in financial markets, multiple independent price paths are generated by using different sequences of random variables. Each simulated path represents a possible realization of future index behaviour. The collection of such paths provides a probabilistic description of future outcomes, allowing for the analysis of expected trends as well as variability.

Furthermore, the aggregation of multiple simulated paths enables the estimation of average price behaviour and the distribution of returns. As the number of simulations increases, the results tend to stabilize, providing a more reliable approximation of the underlying stochastic process.

Thus, the simulation framework translates the theoretical GBM model into a practical computational procedure for generating future index trajectories, forming the basis for subsequent risk analysis and model evaluation.

3.6 Monte Carlo Simulation

Monte Carlo simulation provides a numerical framework for analysing the stochastic behaviour of the NIFTY Bank Index by generating multiple possible future realizations of its price path. The method relies on repeated random sampling to approximate the probabilistic distribution of future outcomes under the Geometric Brownian Motion (GBM) model.

The simulation is performed over a fixed forward time horizon of one year, corresponding to approximately 252 trading days. To capture the variability in future outcomes, a large number of independent simulation paths are generated.

3.6.1 Algorithm

This subsection outlines the step-by-step procedure for implementing Monte Carlo simulation of index prices. It describes the generation of random shocks, iterative price evolution, and repetition across multiple paths. The algorithm provides a structured approach to simulate the stochastic dynamics under the Geometric Brownian Motion model.

Step 1: Data Preparation

Historical daily closing prices of the NIFTY Bank Index are used to compute logarithmic returns. Based on these returns, the drift (μ) and volatility (σ) parameters are estimated as described in Section 3.5.4.

Step 2: Initialization

Set the initial index value $S(0)$ equal to the most recent observed closing price. Define the simulation parameters as:

- Time horizon: $T = 1$ year
- Number of time steps: $N = 252$ (trading days)
- Time increment: $\Delta t = \frac{1}{252}$
- Number of simulation paths: $M = 2000$

These parameters ensure that the simulation reflects realistic market trading conditions over a one-year period.

Step 3: Generation of Random Shocks

For each time step and for each simulation path, independent random variables $\xi(t)$ are generated from a standard normal distribution $N(0,1)$. These random shocks represent the stochastic component of price movements.

Step 4: Price Path Simulation

The index price is simulated iteratively using the discrete GBM formulation:

$$S(t + \Delta t) = S(t) + \mu S(t) \Delta t + \sigma S(t) \xi(t) \sqrt{\Delta t}$$

Starting from the initial value $S(0)$, this equation is applied sequentially over 252 time steps to generate one complete price path. Each realization of the random shocks produces a distinct trajectory.

Step 5: Repetition of Simulation

Steps 3 and 4 are repeated independently for $M = 2000$ simulation paths. This results in a collection of possible future price trajectories, capturing the uncertainty inherent in the market.

Step 6: Aggregation of Results

The ensemble of simulated paths is used to construct the empirical distribution of future prices and returns at each time step. Statistical quantities such as the mean, variance, and percentiles are computed from this distribution. As the number of simulated paths increases, the estimates become more stable and converge toward the expected theoretical behaviour.

Step 7: Output Generation

The simulated price paths provide a probabilistic representation of future index movements. These outputs are subsequently used for analysing return distributions and for computing risk measures such as Value at Risk (VaR) and Conditional Value at Risk (CVaR).

The Monte Carlo simulation framework thus enables a comprehensive exploration of possible future scenarios over a one-year horizon, incorporating both deterministic trends and stochastic fluctuations in the index dynamics.

3.6.2 Implementation

The Monte Carlo simulation is implemented using Python with standard scientific computing libraries. The historical price data is processed using *pandas*, and logarithmic returns are computed to estimate the drift and volatility parameters as described earlier.

The simulation is carried out over a one-year horizon of 252 trading days, with a time step of $\Delta t = \frac{1}{252}$. The initial value is taken as the most recent observed index level. Random shocks are generated using *NumPy* from a standard normal distribution and are used to simulate price paths using the discrete GBM formulation.

A total of 2000 independent simulation paths are generated, producing a set of possible future trajectories of the index. The simulation is implemented using vectorized operations to ensure computational efficiency.

The resulting simulated paths are analysed to obtain summary statistics and are visualized using plotting tools to understand the variability and distribution of future index values.

3.7 Bootstrap Simulation

In addition to the parametric modelling approach based on Geometric Brownian Motion, a non-parametric simulation framework is employed using bootstrap resampling. This approach does not assume any specific probability distribution for returns and instead relies directly on the empirical distribution observed in historical data.

Let $r_1, r_2, \dots, r_n$ denote the historical log returns of the NIFTY Bank Index. The empirical distribution of returns can be represented as:

$$\hat{F}(r) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{r_i \leq r\}}$$

where $\hat{F}(r)$ is the Empirical Cumulative Distribution Function (ECDF), and $\mathbf{1}$ is the indicator function defined in the following way:

$$\mathbf{1} = \begin{cases} 1, & \text{if } r_i \leq r \\ 0, & \text{if } r_i > r \end{cases}$$

Bootstrap simulation involves generating new return sequences by sampling with replacement from the observed returns. Specifically, for each time step t , a return r_t^* is drawn independently from the empirical distribution:

$$r_t^* \sim \hat{F}$$

This process produces a sequence of resampled returns $\{r_1^*, r_2^*, \dots, r_T^*\}$, where $T = 252$ corresponds to the simulation horizon.

Given an initial index value $S(0)$, the simulated price path is constructed using the cumulative product of returns:

$$S(t) = S(0) * \exp \left(\sum_{i=1}^t r_i^* \right)$$

Equivalently, this can be written in recursive form as:

$$S(t+1) = S(t) * \exp(r_i^*)$$

This process is repeated for 252-time steps (one year), and a complete price trajectory is obtained. To capture uncertainty, the simulation is performed for 2000 independent paths, each using a different sequence of randomly sampled returns.

The main advantage of the bootstrap approach is that it preserves important characteristics of real market data, such as skewness, heavy tails, and extreme movements. Since it is based entirely on observed data, it can better reflect actual market behaviour compared to models that assume normality.

However, this method assumes that past return patterns will continue in the future and does not explicitly account for time-varying volatility or changing market conditions.

The simulated price paths obtained from bootstrap sampling are later compared with those from the GBM-based Monte Carlo simulation to evaluate differences in risk and distributional behaviour.

3.8 Risk Measures

The simulated price and return distributions obtained from both the GBM-based Monte Carlo and bootstrap approaches are used to quantify financial risk. In this study, risk is evaluated using two widely adopted measures: Value at Risk (VaR) and Conditional Value at Risk (CVaR). These measures provide insight into potential losses under adverse market conditions.

3.8.1 Value at Risk (VaR)

Value at Risk (VaR) is a statistical measure that estimates the maximum expected loss over a given time horizon at a specified confidence level. It provides a threshold such that losses are unlikely to exceed this value with a chosen level of confidence.

Let R denote the return of the index over the simulation horizon. The VaR at confidence level α is defined as the lower quantile of the return distribution:

$$\text{VaR}_\alpha = \inf \{x \in \mathbb{R} \mid P(R \leq x) \geq 1 - \alpha\}$$

In practical terms, VaR is computed as the $(1-\alpha)$ -quantile of the simulated return distribution. For example, at a 95% confidence level, VaR corresponds to the 5th percentile of returns.

This implies that; With 95% confidence, the loss will not exceed the VaR value over the specified time horizon.

VaR is easy to interpret and widely used in financial risk management. However, it does not provide information about the magnitude of losses beyond the threshold.

3.8.2 Conditional Value at Risk (CVaR)

Conditional Value at Risk (CVaR), also known as Expected Shortfall, addresses the limitation of VaR by measuring the average loss in the worst-case scenarios beyond the VaR threshold.

Mathematically, CVaR at confidence level α is defined as:

$$\text{CVaR} = \mathbb{E} [R \mid R \leq \text{VaR}_\alpha]$$

Thus, CVaR represents the expected return (or loss) given that the loss has already exceeded the VaR level.

In the context of simulated returns, CVaR is computed as the average of the worst $(1 - \alpha)\%$ of outcomes. For example, at a 95% confidence level, CVaR is the average of the lowest 5% of simulated returns.

This measure provides a more comprehensive assessment of downside risk, particularly in the presence of heavy tails and extreme events.

3.8.3 Application to Simulation Results

In this study, VaR and CVaR are computed using the return distributions obtained from both the GBM-based Monte Carlo simulation and the bootstrap simulation. The comparison of these measures across the two approaches provides insight into how different modelling assumptions affect the estimation of risk.

Since the bootstrap method preserves empirical characteristics such as extreme returns, it is expected to produce higher (more conservative) risk estimates compared to the GBM model, which assumes normally distributed returns.

The use of these risk measures enables a quantitative evaluation of potential losses and highlights the importance of modelling assumptions in financial risk analysis.

Chapter 4

Results and Discussion

This chapter presents the results obtained from the stochastic modelling and simulation of the NIFTY Bank Index using both Geometric Brownian Motion (GBM) and bootstrap approaches. The simulated price paths and return distributions are analysed to understand the variability and uncertainty inherent in the index dynamics.

The results from the two modelling frameworks are compared to evaluate their effectiveness in capturing market behaviour. In addition, risk measures such as Value at Risk (VaR) and Conditional Value at Risk (CVaR) are computed and interpreted to assess downside risk. The discussion focuses on drawing meaningful insights from the simulation outcomes and examining the impact of modelling assumptions on the estimation of risk.

4.1 Monte Carlo Simulation (GBM) Results

This section presents the results obtained from simulating the NIFTY Bank Index using the Geometric Brownian Motion (GBM) framework through Monte Carlo methods. The simulation provides a probabilistic view of future index behaviour over a one-year horizon.

Estimated Model Parameters

The GBM model parameters are estimated from historical log return data as described in Section 3.5.4. The estimated drift and volatility are given by:

$$\mu = 0.000425, \quad \sigma = 0.013723$$

The drift parameter represents the average expected return of the index, while the volatility parameter captures the uncertainty or variability in returns. A relatively higher value of volatility indicates greater dispersion in future price trajectories.

Simulated Price Paths

The simulated price paths generated using the GBM model are shown in Figures 4.1 and 4.2. Figure 4.1 illustrates a subset of simulated paths to provide visual clarity, while Figure 4.2 presents all 2000 simulated trajectories.

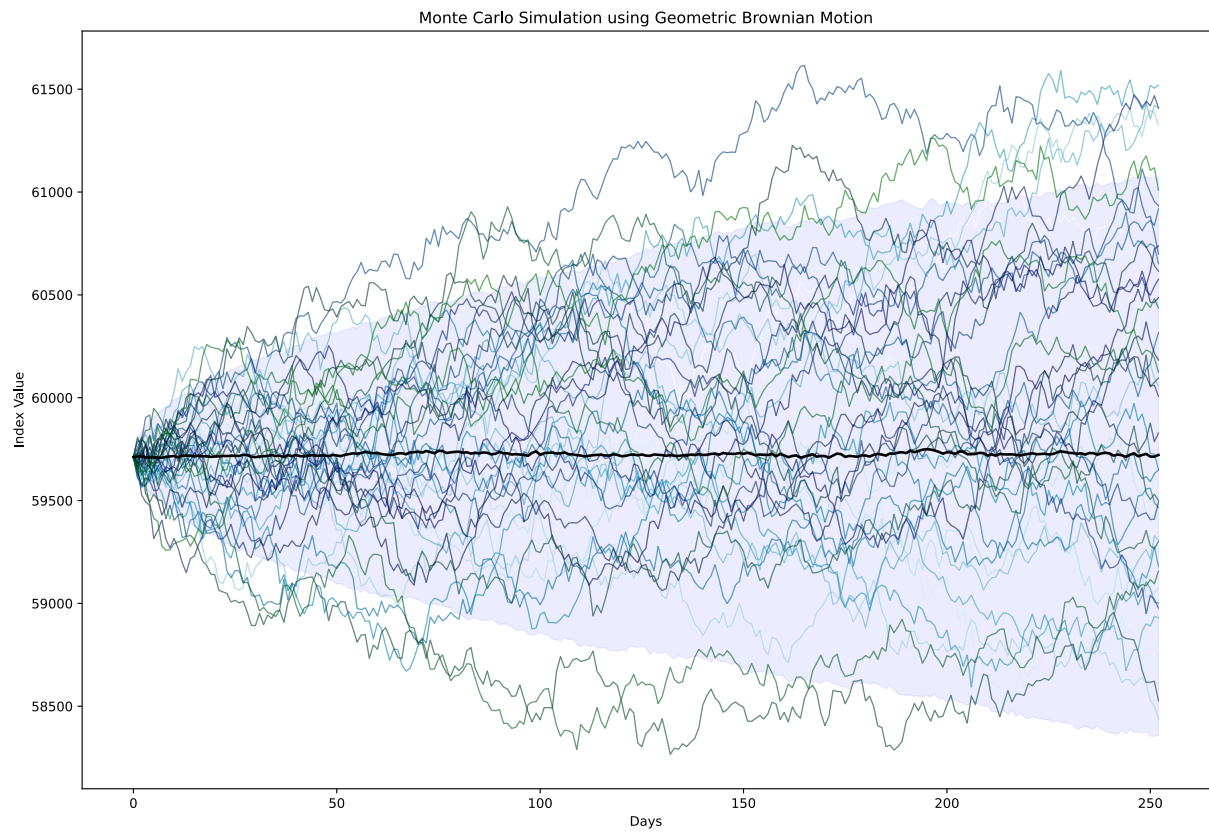


Figure 4.1: Monte Carlo Simulation Chart Showing 40 Price Paths

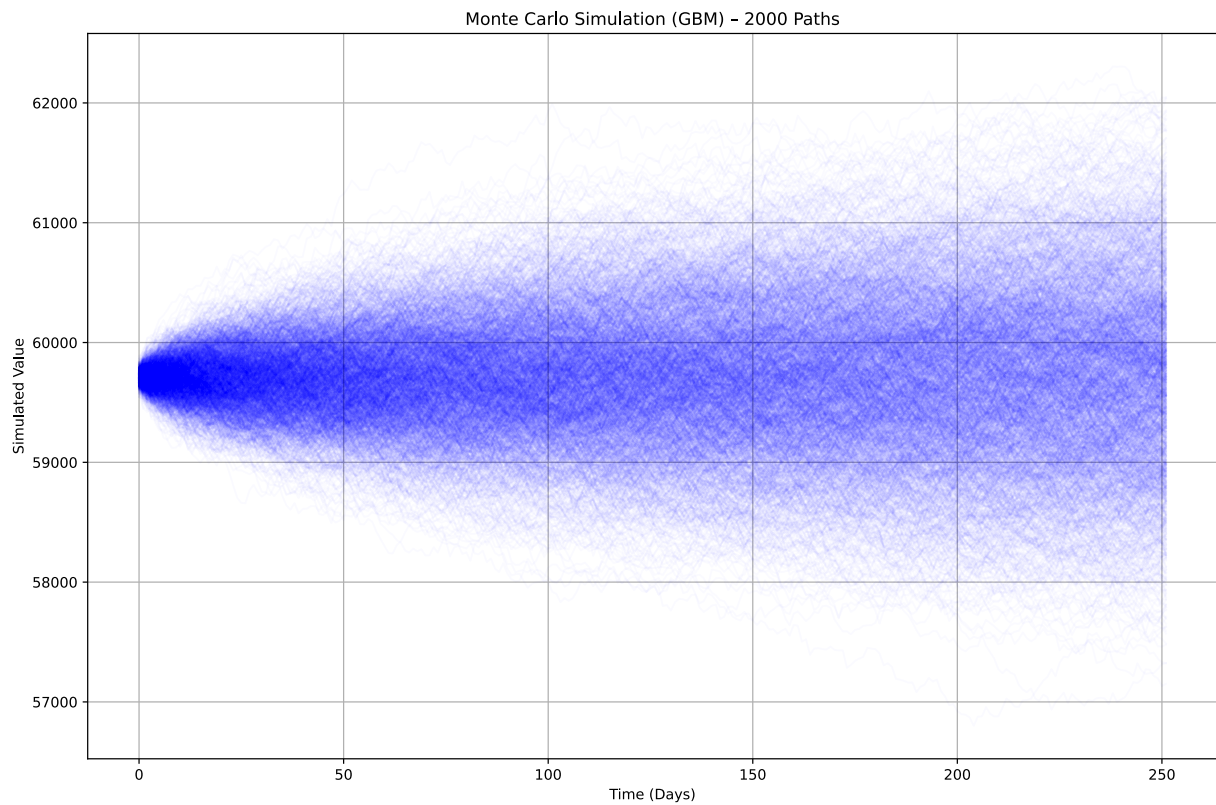


Figure 4.2: Monte Carlo Simulation Chart Showing 2000 Price Paths.

The simulated paths exhibit significant variability, with different trajectories diverging over time despite starting from the same initial value. This divergence reflects the stochastic nature of the model, where random shocks at each time step influence the evolution of prices.

A key observation is that the spread of the paths increases as time progresses. This indicates that uncertainty accumulates over time, leading to a wider range of possible future outcomes. The overall behaviour remains smooth and continuous, consistent with the assumptions of GBM, which does not account for abrupt jumps or discontinuities in price movements.

Mean Path and Confidence Bands

The average simulated price trajectory along with confidence bands is shown in Figure 4.3.

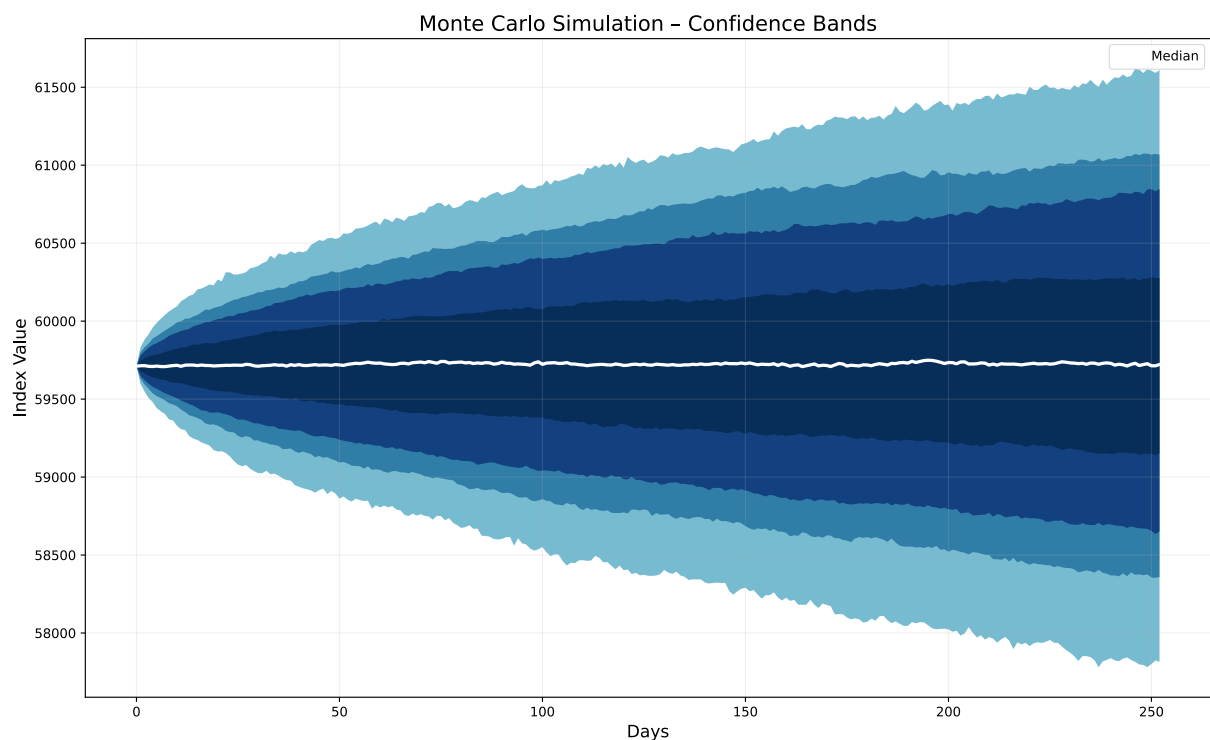


Figure 4.3: Monte Carlo Simulation chart showing the mean price path and Confidence bands

The mean path represents the expected evolution of the index under the GBM model, while the Lighter bands to Darker bands correspond to (1st - 99th), (5th - 95th), (10th - 90th), & (25th - 75th) percentiles respectively capture the range within which most simulated paths lie.

It is observed that the confidence bands widen over time, indicating increasing dispersion in simulated prices. This behaviour is consistent with the theoretical properties of GBM, where the variance of the process grows with time. The mean trajectory shows a gradual trend governed by the estimated drift parameter, while the bands reflect the influence of volatility.

These confidence intervals provide a useful representation of uncertainty, highlighting the range of likely outcomes rather than a single deterministic prediction.

Distribution of Terminal Prices

The distribution of simulated index values at the end of the one-year horizon is presented in Figure 4.4. The histogram exhibits a right-skewed shape, which is characteristic of the log-normal distribution implied by the GBM model.

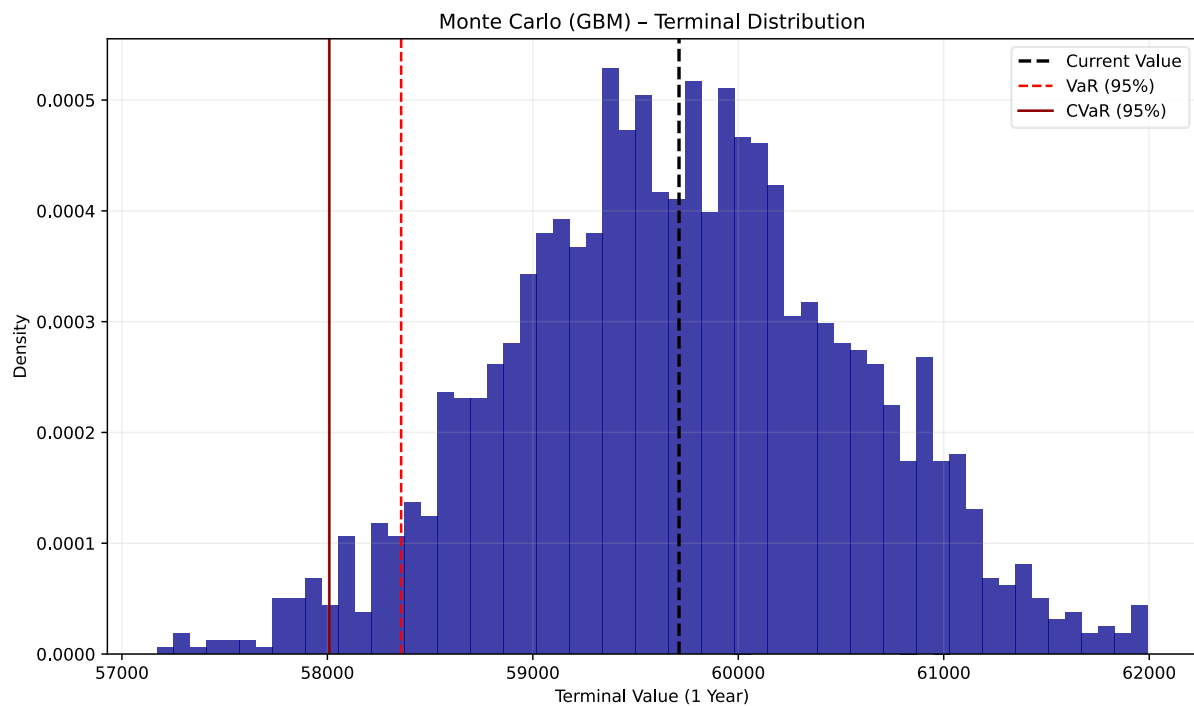


Figure 4.4: Terminal Distribution Prices at end of 1 year using GBM

This skewness indicates that while most outcomes cluster around moderate values, there is a non-negligible probability of higher index levels. At the same time, the lower tail is relatively limited, reflecting the model's assumption of normally distributed returns. The terminal

distribution provides a summary of all simulated paths and forms the basis for further risk analysis in subsequent sections.

Discussion of GBM Behaviour

The GBM-based Monte Carlo simulation captures key aspects of stochastic price evolution, including randomness, continuous trajectories, and increasing uncertainty over time. However, the model exhibits certain limitations. In particular, the assumption of normally distributed returns leads to relatively thin tails in the distribution, which may underestimate the likelihood of extreme market movements.

These observations highlight the importance of comparing GBM results with alternative approaches, such as bootstrap simulation, which can better capture empirical features of financial data.

4.2 Bootstrap Simulation Results

This section presents the results obtained from the bootstrap-based simulation of the NIFTY Bank Index. Unlike the GBM framework, which assumes normally distributed returns, the bootstrap method generates future scenarios by resampling directly from historical log returns. This allows the simulation to preserve empirical characteristics such as skewness, heavy tails, and extreme movements.

The simulation is performed over a one-year horizon of 252 trading days, with 2000 independent bootstrap paths generated using resampled return sequences.

Simulated Price Paths

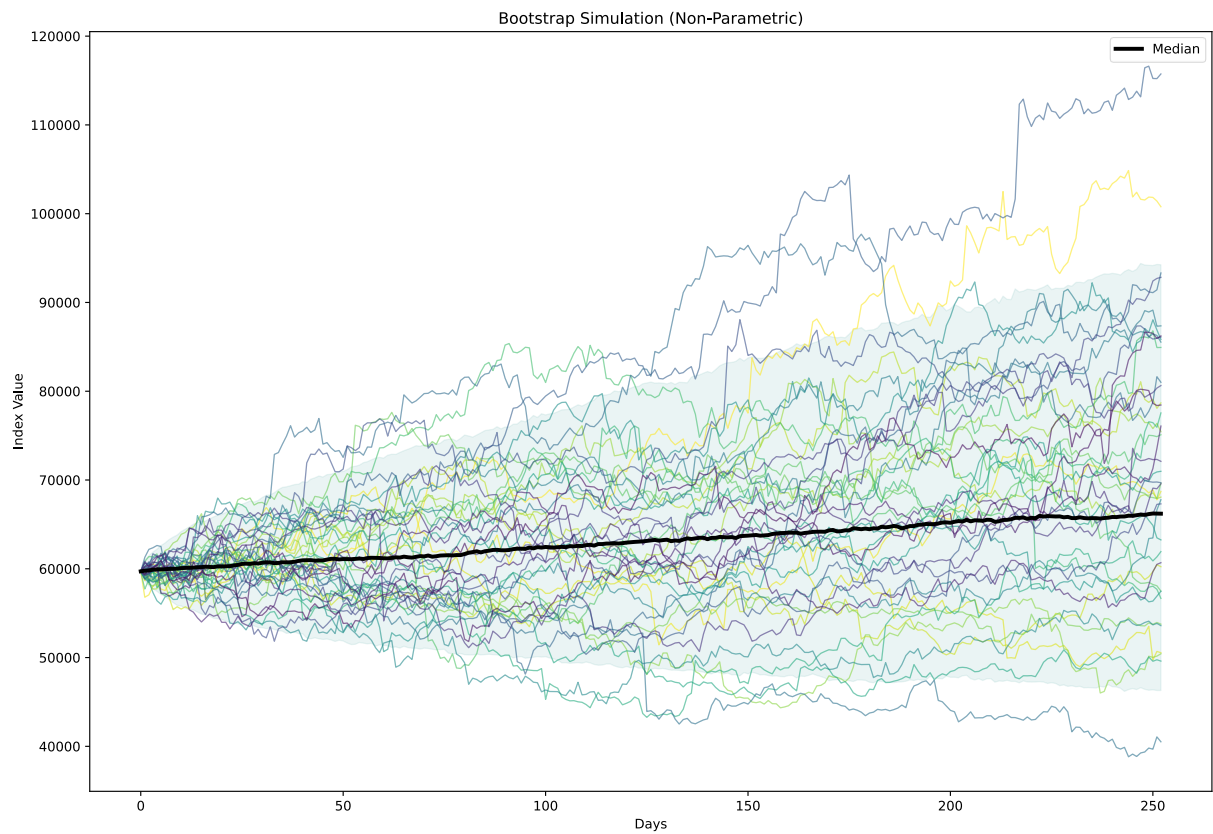


Figure 4.5: Bootstrap Simulation showing 40 price paths

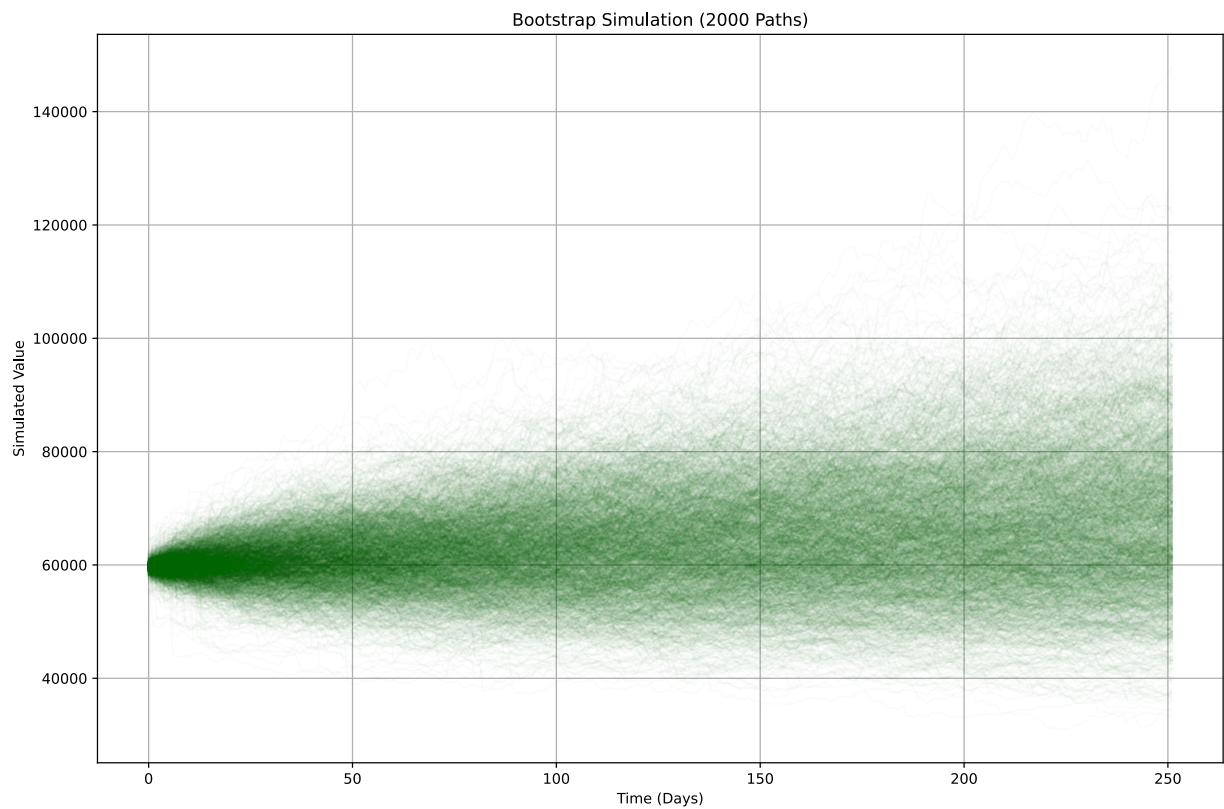


Figure 4.6: Bootstrap Simulation chart showing 2000 price paths

The simulated price paths obtained from the bootstrap method are shown in Figures 4.5 and 4.6. Figure 4.5 presents a subset of paths for visual clarity, while Figure 4.6 displays all 2000 simulated trajectories.

Compared to the GBM-based simulation, the bootstrap paths exhibit greater irregularity and variability. The trajectories reflect abrupt changes and fluctuations, which arise from the direct use of historical returns rather than smooth normally distributed shocks. This results in more realistic price movements that resemble observed market behaviour.

The dispersion of paths increases over time, similar to the GBM case, indicating growing uncertainty. However, the spread appears less uniform and more influenced by extreme values present in the historical data.

Mean Path and Confidence Bands

The average simulated trajectory along with confidence bands is shown in Figure 4.7.

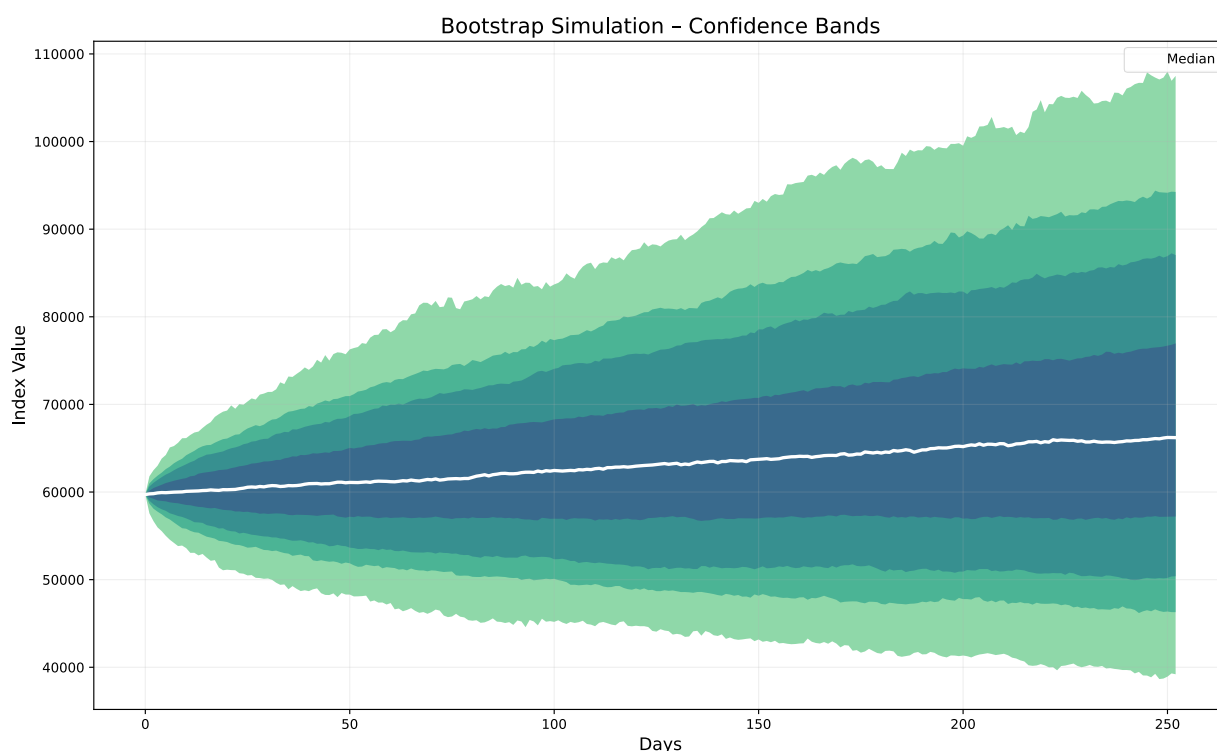


Figure 4.7: Bootstrap Simulation chart showing the mean price path and Confidence bands

The mean path represents the expected evolution of the index under the bootstrap framework, while the bands capture the variability of simulated outcomes.

The confidence bands in the bootstrap simulation are observed to be wider and less symmetric compared to those of the GBM model. This reflects the presence of skewness and extreme returns in the empirical data. The widening of the bands over time again indicates increasing uncertainty, but with a stronger influence of tail behaviour.

Distribution of Terminal Prices

The distribution of simulated index values at the end of the one-year horizon is shown in Figure 4.8. Unlike the GBM case, the terminal distribution obtained from bootstrap simulation may exhibit deviations from the ideal log-normal shape.

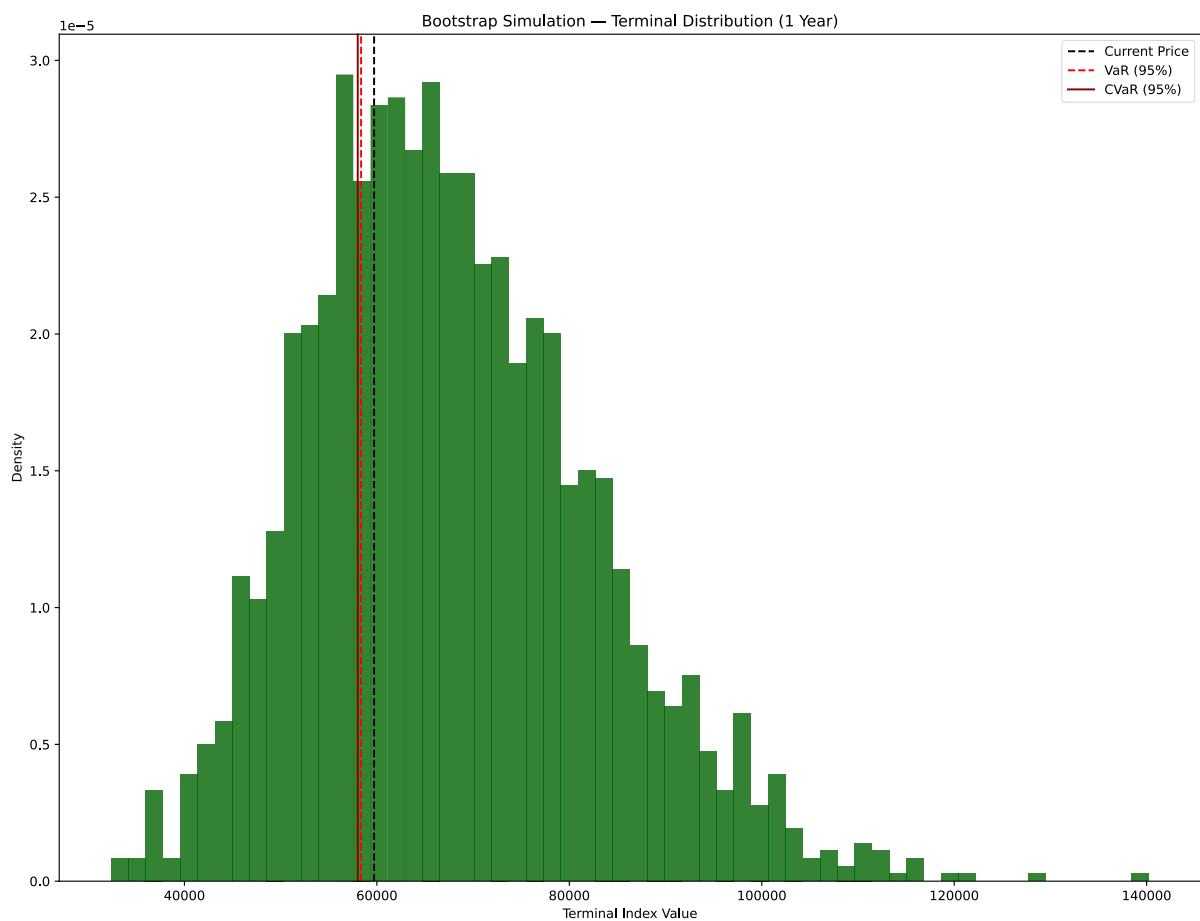


Figure 4.8: Terminal Distribution of prices at the end of one year using Bootstrap Simulation

The histogram typically shows heavier tails and possible asymmetry, indicating a higher probability of extreme outcomes. This behaviour arises from the fact that the bootstrap method directly incorporates observed large positive and negative returns from the historical dataset.

The terminal distribution thus provides a more realistic representation of potential future outcomes, especially in terms of extreme scenarios.

Discussion of Bootstrap Behaviour

The bootstrap simulation captures important empirical features of financial time series that are not fully represented in parametric models such as GBM. In particular, it reflects heavy tails, skewness, and irregular fluctuations in price paths.

However, the method relies entirely on historical data and assumes that past return patterns will persist in the future. It does not explicitly model time-varying volatility or structural changes in market conditions.

Despite these limitations, the bootstrap approach serves as a valuable complement to the GBM model, providing a data-driven perspective on uncertainty and risk. The differences observed between the two approaches are further analysed in the subsequent risk analysis section.

4.3 Risk Analysis and Model Comparison

This section presents the risk estimates obtained from the simulated return distributions of both the GBM-based Monte Carlo model and the bootstrap simulation. The risk is quantified using Value at Risk (VaR) and Conditional Value at Risk (CVaR) at the 95% confidence level.

Value at Risk (VaR)

The estimated VaR values at the 95% confidence level are:

Historical VaR: 0.019500

GBM VaR: 0.022896

Bootstrap VaR: 0.254610

The GBM-based VaR is slightly higher than the historical estimate, indicating a moderate level of risk captured by the model. In contrast, the bootstrap VaR is significantly larger, reflecting the presence of extreme negative returns in the empirical data. This suggests that the bootstrap approach captures tail risk more aggressively compared to the GBM model.

Conditional Value at Risk (CVaR)

The estimated CVaR values at the 95% confidence level are:

Historical CVaR: 0.032783

GBM CVaR: 0.028898

Bootstrap CVaR: 0.352215

The GBM-based CVaR is slightly lower than the historical estimate, indicating that the model underestimates extreme losses due to its assumption of normally distributed returns. On the other hand, the bootstrap CVaR is substantially higher, capturing the severity of losses in the worst-case scenarios.

Model Comparison

The GBM model underestimates tail risk due to its normality assumption. The bootstrap method captures extreme events, yielding more realistic risk estimates.

Chapter 5

Summary and Conclusions

Summary and Key Findings

This study presented a comprehensive stochastic modelling framework for analysing the future behaviour and risk characteristics of the NIFTY Bank Index using both parametric and non-parametric approaches. The analysis was based on historical daily price data, which was transformed into logarithmic returns to enable probabilistic modelling.

An exploratory data analysis revealed important empirical features of the index, including non-normal return distributions, volatility clustering, and the presence of extreme market movements. These observations highlighted the limitations of simple modelling assumptions and motivated the use of both Geometric Brownian Motion (GBM) and bootstrap simulation techniques.

Under the GBM framework, model parameters such as drift and volatility were estimated from historical returns, and Monte Carlo simulation was used to generate multiple future price paths. The resulting trajectories exhibited smooth behaviour with increasing dispersion over time, consistent with the theoretical properties of GBM. However, the simulated distributions were relatively symmetric and exhibited thinner tails.

In contrast, the bootstrap simulation approach generated future scenarios by resampling directly from historical returns. This method preserved empirical characteristics such as skewness and heavy tails, resulting in more irregular price paths and wider distributions. The bootstrap-based simulations reflected extreme movements more prominently, indicating a higher level of variability.

The comparison of risk measures further reinforced these observations. The GBM model produced moderate estimates of Value at Risk (VaR) and Conditional Value at Risk (CVaR), while the bootstrap approach yielded significantly higher values, highlighting its sensitivity to extreme outcomes. These findings demonstrate the importance of modelling assumptions in accurately capturing financial risk.

Conclusions and Insights

The results of this study emphasize that stochastic modelling provides a powerful framework for understanding uncertainty in financial markets. While the GBM model offers analytical simplicity and tractability, its reliance on constant volatility and normally distributed returns limits its ability to capture real-world market behaviour.

The bootstrap approach, being entirely data-driven, provides a more realistic representation of market dynamics by incorporating observed irregularities and extreme events. This makes it particularly suitable for risk analysis, where accurate estimation of tail behaviour is crucial.

A key insight from this study is that no single model can fully capture the complexity of financial markets. Instead, the choice of model should depend on the objective of the analysis. For general trend estimation and theoretical modelling, GBM remains useful. However, for risk-sensitive applications, especially those involving extreme losses, non-parametric approaches such as bootstrap simulation provide more reliable estimates.

Overall, the study highlights the critical role of distributional assumptions in financial modelling and underscores the need to complement theoretical models with data-driven methods for a more comprehensive understanding of market uncertainty.

Limitations and Future Scope

Despite providing valuable insights, the study is subject to certain limitations. The GBM model assumes constant drift and volatility, which does not reflect the time-varying nature of real financial markets. It also fails to account for sudden jumps or structural changes in market behaviour. On the other hand, the bootstrap approach relies entirely on historical data and assumes that past patterns will persist in the future, which may not always hold true.

Furthermore, the models considered in this study do not explicitly incorporate macroeconomic factors, market sentiment, or external shocks that can significantly influence price movements. The assumption of independence in return sampling may also overlook temporal dependencies present in financial time series.

Future research can address these limitations by incorporating more advanced models that account for time-varying volatility, such as GARCH models, or by considering regime-switching frameworks that capture different market conditions. Additionally, integrating machine learning techniques with stochastic models may improve predictive performance and risk estimation. Hybrid approaches that combine parametric and non-parametric methods could further enhance the robustness of financial modelling.

Appendix

Python Implementation of Stochastic Modelling of NIFTY BANK Index

1. Setup and Imports

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_squared_error
```

2. Load the Dataset

```
# Load the dataset
import pandas as pd

# Replace with your actual dataset path if needed
file_path = "niftybank2015-2026.csv"

# Read the CSV file
df = pd.read_csv(file_path)

# Convert Date column to datetime
df['Date'] = pd.to_datetime(df['Date'], format="%d-%b-%y")

# Sort again just to ensure correct timeline
df = df.sort_values(by="Date")
df.reset_index(drop=True, inplace=True)

# Sort again just to ensure correct timeline
df = df.sort_values(by="Date")
df.reset_index(drop=True, inplace=True)

# Display the first few rows
df.head()
```

3. Quick data overview

```
# Shape of the dataset
print("Dataset Shape:", df.shape)

# Column names
print("Columns:", df.columns.tolist())

# Basic info
df.info()

# Summary statistics
df.describe()
```

4. Cleaning and preprocessing

```
# Check for missing values
print("Missing Values:\n", df.isnull().sum())

# Drop rows with all values missing
df = df.dropna(how="all")

# Remove duplicate rows
df = df.drop_duplicates()

# Sort by date
df = df.sort_values(by="Date").reset_index(drop=True)

# Convert numeric columns to proper data types
numeric_cols = ['Open', 'High', 'Low', 'Close', 'Shares Traded', 'Turnover (INR Cr)']

for col in numeric_cols:
    df[col] = pd.to_numeric(df[col], errors='coerce')

# Handle any remaining missing numeric values by interpolation
df[numeric_cols] = df[numeric_cols].interpolate()

print("Cleaning & preprocessing complete!")
df.head()
```

5. Histograms of the dataset

```
df.hist(figsize=(28.6, 20), bins=30)
plt.tight_layout()
plt.savefig("Histograms.svg", bbox_inches="tight")
plt.show()
```

6. Historical close price of NIFTY BANK Index

```
plt.figure(figsize=(13.5,10))
plt.plot(df['Date'], df['Close'])
plt.xlabel("Date")
plt.ylabel("Close Price")
plt.title("Historical Closing Price of NIFTY BANK Index")
plt.savefig("Historical_Close.svg", bbox_inches="tight")
plt.show()
```

7. Daily Price Range over time

```
df['Price Range'] = df['High'] - df['Low']

plt.figure(figsize=(12, 5))
plt.plot(df['Date'], df['Price Range'])
plt.xlabel("Date")
plt.ylabel("Daily Price Range")
plt.title("Daily Price Range Over Time")
plt.savefig("Daily_Price_Range.svg", bbox_inches="tight")
plt.show()
```

8. Trading Volume over time

```
plt.figure(figsize=(12, 5))
plt.plot(df['Date'], df['Shares Traded'])
plt.xlabel("Date")
plt.ylabel("Shares Traded")
plt.title("Trading Volume Over Time")
plt.savefig("Trading_Volume.svg", bbox_inches="tight")
plt.show()
```

9. Distribution of Daily Returns

```
df['Returns'] = df['Close'].pct_change()
df['Returns'].dropna(inplace=True)
```

```
plt.figure(figsize=(14.3, 10))
plt.hist(df['Returns'], bins=50)
plt.xlabel("Daily Returns")
plt.ylabel("Frequency")
plt.title("Distribution of Daily Returns")
plt.savefig("Daily_Returns.svg", bbox_inches="tight")
plt.show()
```

10. 30 Day Rolling Volatility

```
df['Rolling Volatility'] = df['Returns'].rolling(window=30).std()

plt.figure(figsize=(14.3, 9))
plt.plot(df['Date'], df['Rolling Volatility'])
plt.xlabel("Date")
plt.ylabel("Rolling Volatility (30-day)")
plt.title("30-Day Rolling Volatility")
plt.savefig("30_Day_Rolling_Volatility.svg", bbox_inches="tight")
plt.show()
```

11. Correlation Map

```
plt.figure(figsize=(8, 6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="rocket_r")
plt.title("Correlation Heatmap")
plt.savefig("Correlation_Map.svg", bbox_inches="tight")
plt.show()
```

12. Monte Carlo Simulation using GBM

Parameter Estimation

```
df['LogReturns'] = np.log(df['Close'] / df['Close'].shift(1))
df = df.dropna(subset=['LogReturns']).reset_index(drop=True)
rets = df['LogReturns'].values
mu = np.mean(rets)
sigma = np.std(rets, ddof=1)
S0 = df['Close'].iloc[-1]

print(f"Estimated Parameters:")
print(f"Initial Value (S0): {S0:.2f}")
print(f"Mean (mu): {mu:.6f}")
print(f"Volatility (sigma): {sigma:.6f}")
```

Simulation Plots and Terminal Distribution

```

SEED = 42
np.random.seed(SEED)
N_SIMS = 2000
HORIZON = 252          # days
PLOT_PATHS = 40        # sample paths to display

df['LogReturns'] = np.log(df['Close'] / df['Close'].shift(1))
df = df.dropna(subset=['LogReturns']).reset_index(drop=True)
rets = df['LogReturns'].values
S0 = df['Close'].iloc[-1]
mu = rets.mean(); sigma = rets.std(ddof=1)

Z = np.random.normal(size=(HORIZON, N_SIMS))
dt = 1/252
increments = (mu - 0.5 * sigma**2) * dt + sigma * np.sqrt(dt) * Z
log_paths = np.vstack([np.zeros(N_SIMS), np.cumsum(increments, axis=0)])
S_gbm = S0 * np.exp(log_paths)          # shape (HORIZON+1, N_SIMS)
term_gbm = S_gbm[-1, :]

def var_cvar(terminals, S0, alpha=0.95):
    losses = -np.log(terminals / S0)
    VaR = np.percentile(losses, alpha*100)
    CVaR = losses[losses >= VaR].mean()
    return VaR, CVaR

VaR_g, CVaR_g = var_cvar(term_gbm, S0)

print(f"S0={S0:.2f} mu={mu:.6f} sigma={sigma:.6f}")
print(f"GBM VaR95={VaR_g:.6f}, CVaR95={CVaR_g:.6f}")

p = np.percentile(S_gbm, [5,25,50,75,95], axis=1)
x = np.arange(HORIZON+1)
plt.figure(figsize=(13,9))
cmap = plt.cm.ocean_r

for i in range(min(PLOT_PATHS, N_SIMS)):
    plt.plot(x, S_gbm[:, i], color=cmap(i / PLOT_PATHS), alpha=0.6, linewidth=1)

# Median & confidence bands
plt.plot(x, p[2], color='black', linewidth=2, label='Median')
plt.fill_between(x, p[0], p[4], alpha=0.08, color='blue')

plt.title("Monte Carlo Simulation using Geometric Brownian Motion")
plt.xlabel("Days")
plt.ylabel("Index Value")
plt.tight_layout()
plt.savefig("Monte_Carlo_GBM_Chart.svg", bbox_inches="tight")
plt.show()

HORIZON = S_gbm.shape[0] - 1
x = np.arange(HORIZON + 1)
percentiles = [1, 5, 10, 25, 50, 75, 90, 95, 99]
p = np.percentile(S_gbm, percentiles, axis=1)

plt.figure(figsize=(13, 8))
cmap = plt.cm.ocean_r
bands = [
    (1, 99),
    (5, 95),
    (10, 90),
    (25, 75)
]

for i, (low, high) in enumerate(bands):
    color = cmap(0.3 + 0.15 * i)
    lower = np.percentile(S_gbm, low, axis=1)
    upper = np.percentile(S_gbm, high, axis=1)

    plt.fill_between(x, lower, upper, color=color, alpha=0.6, linewidth=0)

```

```
# Median line
median = np.percentile(S_gbm, 50, axis=1)
plt.plot(x, median, color='white', linewidth=2.5, label='Median')
plt.title("Monte Carlo Simulation - Confidence Bands", fontsize=16)
plt.xlabel("Days", fontsize=13)
plt.ylabel("Index Value", fontsize=13)
plt.grid(alpha=0.2)
plt.legend()
plt.tight_layout()
plt.savefig("Monte_Carlo_GBM_Bands.svg", bbox_inches="tight")
plt.show()
```

13. Bootstrap Simulation Plots, Terminal distribution

```
# Parameters
N_SIMS = 2000
HORIZON = 252      # 1-year forecast
SEED = 42
np.random.seed(SEED)

term_boot = np.empty(N_SIMS)
for i in range(N_SIMS):
    samp = np.random.choice(rets, size=HORIZON, replace=True)
    term_boot[i] = S0 * np.exp(samp.sum())

PLOT_PATHS = 40
df['LogReturns'] = np.log(df['Close'] / df['Close'].shift(1))
df = df.dropna(subset=['LogReturns']).reset_index(drop=True)
rets = df['LogReturns'].values
S0 = df['Close'].iloc[-1]
idx = np.random.randint(0, len(rets), size=(HORIZON, N_SIMS))
boot_samples = rets[idx] # shape: (HORIZON, N_SIMS)
log_paths = np.vstack([np.zeros(N_SIMS), np.cumsum(boot_samples, axis=0)])
S_boot = S0 * np.exp(log_paths) # shape: (HORIZON+1, N_SIMS)
term_boot = S_boot[-1, :]

p = np.percentile(S_boot, [5, 25, 50, 75, 95], axis=1)
x = np.arange(HORIZON + 1)
plt.figure(figsize=(13, 9))
cmap = plt.cm.viridis_r

for i in range(min(PLOT_PATHS, N_SIMS)):
    plt.plot(x, S_boot[:, i], color=cmap(i / PLOT_PATHS), alpha=0.6, linewidth=1)

# median
plt.plot(x, p[2], color='black', linewidth=3, label='Median')

# confidence band
plt.fill_between(x, p[0], p[4], alpha=0.1, color=cmap(0.5))
plt.title("Bootstrap Simulation (Non-Parametric)")
plt.xlabel("Days")
plt.ylabel("Index Value")
plt.legend()
plt.tight_layout()
plt.savefig("Bootstrap_Simulation_Chart.svg", bbox_inches="tight")
plt.show()

import numpy as np
import matplotlib.pyplot as plt

HORIZON = S_boot.shape[0] - 1
x = np.arange(HORIZON + 1)
percentiles = [1, 5, 10, 25, 50, 75, 90, 95, 99]
p = np.percentile(S_boot, percentiles, axis=1)
plt.figure(figsize=(13, 8))
cmap = plt.cm.viridis_r
bands = [
    (1, 99),
    (5, 95),
    (10, 90),
    (25, 75)
]
```

```

# Normalize colors
for i, (low, high) in enumerate(bands):
    color = cmap(0.3 + 0.15 * i)
    lower = np.percentile(S_boot, low, axis=1)
    upper = np.percentile(S_boot, high, axis=1)

    plt.fill_between(x, lower, upper, color=color, alpha=0.6, linewidth=0)

# Median line
median = np.percentile(S_boot, 50, axis=1)
plt.plot(x, median, color='white', linewidth=2.5, label='Median')

plt.title("Bootstrap Simulation - Confidence Bands", fontsize=16)
plt.xlabel("Days", fontsize=13)
plt.ylabel("Index Value", fontsize=13)

plt.grid(alpha=0.2)
plt.legend()

plt.tight_layout()
plt.savefig("Bootstrap_Simulation_Bands.svg", bbox_inches="tight")
plt.show()

plt.figure(figsize=(13, 10))
plt.hist(term_boot, bins=60, density=True, alpha=0.8, color='darkgreen')
plt.axvline(S0, color='black', linestyle='--', label='Current Price')
plt.xlabel("Terminal Index Value")
plt.ylabel("Density")
plt.title("Bootstrap Simulation - Terminal Distribution (1 Year)")
# VaR & CVaR lines
plt.axvline(S0 * np.exp(-VaR_g), color='red', linestyle='--', label='VaR (95%)')
plt.axvline(S0 * np.exp(-CVaR_g), color='darkred', linestyle='-', label='CVaR (95%)')

plt.legend()
plt.tight_layout()
plt.savefig("Bootstrap_Terminal_Distribution.svg", bbox_inches="tight")
plt.show()

```

14. Risk Metrics VaR and CVaR

```

print(f"Historical VaR (95%): {hist_VaR_95:.6f}")
print(f"GBM VaR (95%):      {VaR_g:.6f}")
print(f"Bootstrap VaR (95%):  {VaR_b:.6f}")

print(f"Historical CVaR (95%): {hist_CVaR_95:.6f}")
print(f"GBM CVaR (95%):       {CVaR_g:.6f}")
print(f"Bootstrap CVaR (95%):   {CVaR_b:.6f}")

```


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